DIFFEV

Users Guide

Version 6.23

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Preface

Disclaimer

The DIFFEV software described in this guide is provided without warranty of any kind. No liability is taken for any loss or damages, direct or indirect, that may result through the use of *DIFFEV*. No warranty is made with respect to this manual, or the program and functions therein. There are no warranties that the programs are free of error, or that they are consistent with any standard, or that they will meet the requirement for a particular application. The programs and the manual have been thoroughly checked. Nevertheless, it can not be guaranteed that the manual is correct and up to date in every detail. This manual and the DIFFEV program may be changed without notice.
DIFFEV is intended as a public domain program. It may be used free of charge. Any commercial use is, however, not allowed without permission of the authors.

Using DIFFEV

Publications of results totally or partially obtained using the program DIFFEV should state that DIFFEV was used and contain the following reference:

> NEDER, R.B. in preparation - check website.

More information

This users guide can only provide program specific details. A broader discussion of simulation techniques and some DIFFEV examples and macro files can be found in our book

> NEDER, R.B. & PROFFEN, TH. "Diffuse Scattering and Defect Structure Simulations - A cook book using the programs DISCUS", *IUCr Texts on Crystallography*, Oxford University Press, 2007.

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Chapter 1

Introduction

1.1 What is DIFFEV ?

DIFFEVis a generic evolutionary refinement program that implements the differential evolutionary algorithm Price et al. (2005). Evolutionary or genetic refinement algorithms allow the refinement of models, functions, or more generally speaking the parameters of a cost function to obtain a good solution.
A least squares based refinement of a function $y = F(p_0, p_1, ..., p_n)$ requires the calculation of all partial derivatives $\partial y/\partial p_i$, either from an analytical or a numeric solution. If these derivatives cannot be calculated, either because they cannot be derived analytically or because the numeric computation is too time consuming, evolutionary algorithms offer a possibility to refine these parameters. All algorithms are population based, i.e. several different parameter sets $P_I : [p_0, p_1, ..., p_n]$ are created simultaneously. For each of these parameter sets, the value of the cost function is computed. In the next step, a new group of parameter sets is generated and the cost function calculated anew. The respective values of the cost function are compared and those parameter sets that yield the better cost function will in turn be taken to generate the next generation of parameter sets. By carefully designed modification of the parameter values from generation to generation and by weeding out those parameter sets that lead to a bad fit, the algorithm will eventually find parameter sets that provide a good fit to the experimentally determined function.
DIFFEV provides the refinement part of such an evolutionary algorithm. It creates the group of parameter sets, compares the cost function values between two successive generations and creates the next generation based on a comparison between the old and new cost function values. It does, however, not calculate the cost function itself. This task is handed over to a slave program. Since this slave program could calculate any cost function, DIFFEV is not limited to the refinement of a particular physical problem.

1.2 What is new ?

DIFFEV is available as a stand alone programmed and may also be used within the DISCUS SUITE. The DISCUS SUITE is optimized with respect to the performance on a large scale computing facility. Several new features are available within the DISCUS SUITE. These are explained

in the DISCUS SUITE manual.

1.3 Getting started

After the program DISCUS SUITE is installed properly and the environment variables are set, the program can be started by typing 'discus_suite' at the operating systems prompt for a Linux or MACos operating system. For a Windows operating ssytem, double click on the `discus_suite` icon on your desktop.

Symbol	Description
"text"	Text given in double quotes is to be understood as typed.
<text>	Text given in angled brackets is to be replaced by an appropriate value, if the corresponding line is used in DIFFEV. It could, for example be the actual name of a file, or a numerical value.
`text`	Text in single quotes exclusively refers to DIFFEV commands.
[text]	Text in square brackets describes an optional parameter or command. If omitted, a default value is used, else the complete text given in the square brackets is to be typed.
{text | text}	Text given in curly brackets is a list of alternative parameters. A vertical line separates two alternative, mutually exclusive parameters.

Table 1.1: Used symbols

The program uses a command language to interact with the user. The command `exit` terminates the program and returns control to the shell. All commands of DIFFEV consist of a command verb, optionally followed by one or more parameters. All parameters must be separated from one another by a comma ",". There is no predefined need for any specific sequence of commands. DIFFEV is case sensitive, all commands and alphabetic parameters MUST be typed in lower case letters. If DIFFEV has been compiled using the `-DREADLINE` option (see installation files) basic line editing and recall of commands is possible. For more information refer to the reference manual or check the online help using (`help command input`). Names of input or output files are to be typed as they will be expected by the shell. If necessary include a path to the file. All commands may be abbreviated to the shortest unique possibility. At least a single space is needed between the command verb and the first parameter. No comma is to precede the first parameter. A line can be marked as comment by inserting a "`#`" as first character in the line.
The symbols used throughout this manual to describe commands, command parameters, or explicit text used by the program DIFFEV are listed in Table 1.1. There are several sources of information, first DIFFEV has a build in online help, which can be accessed by entering the command `help` or if help for a particular command <cmd> is wanted by `help <cmd>`. This manual describes background and principle functions of DIFFEV and should give some insight in the ways to use this program.
DIFFEV is distributed as part of the diffuse scattering simulation software DISCUS. However, DIFFEV can be used as general refinement program separate from the DISCUS program package.

Variable	Description
pop_gen[1]	Current population number
pop_n[1]	Number of members in the population
pop_c[1]	Number of children in the population
pop_dimx[1]	Number of parameters to be refined
diff_cr[1]	Cross over probability
diff_f[1]	Scale factor for the difference vectors
diff_k[1]	point along line between parent and donor base
diff_lo[1]	Local search probability
diff_sel[1] *	Selection mode: 0 compare to parent; 1 use best of (members and children); 2 use best of (children)
pop_xmin[i]	Minimum allowed value for parameter no. i
pop_xmax[i]	Maximum allowed value for parameter no. i
pop_smin[i]	Minimum allowed starting value for parameter no. i
pop_smax[i]	Maximum allowed starting value for parameter no. i
pop_sig[i]	Global sigma for parameter no. i
pop_lsig[i]	Local search sigma for parameter no. i
pop_v[i,j] *	Value of parameter no. i for member no. j
pop_t[i,j] *	Value of current trial parameter no. i for child no. j
rvalue[i] *	R-value for member no. i
child_val[i]	R-value for child no. i
bestm[1] *	Number of member with best R-value
bestr[1] *	Best R-value
worstm[1] *	Number of member with worst R-value
worstr[1] *	Worst R-value

Table 1.2: DIFFEV variables. Variables marked with * are read-only and cannot be altered.

1.4 Command language

The program includes a FORTRAN style interpreter that allows the user to program complex modifications. A detailed discussion about the command language which is common to all DISCUS package programs can be found in the separate DISCUS package reference guide which is included with the package. Table 1.2 shows a summary of DIFFEV specific variables. Some of these variables can not be modified, others like can be altered, thus allowing to modify the refinement strategy.

Chapter 2

Differential Evolutionary Algorithm

2.1 Refinement via evolutionary algorithms

Every time we measure some physical effect and wish to understand how this effect works, we want to determine the parameters of a model function that will replicate the observations. The term refinement refers to the process by which the parameters of the function are tuned such as to give the best agreement between the observed and calculated values. The term *best agreement* merits careful definition, for right now it is sufficient to say that the sum over all squared differences between the observations and the calculations shall be minimized. Thus, refinement is but a special case of general optimization. A very different example for an optimization could be the task to place as many integrated circuits into a chip and simultaneously achieve the fastest computations. Quite well known is the traveling salesman problem. Here the optimization task requires to find the shortest route that visits a number of spots distributed in space.
By far the fastest refinement technique is a least squares algorithm. Such an algorithm can always be applied if we can describe the physical effect as a function of parameters:

$$y \; = \; F(p_0, p_1, ..., p_n), \tag{2.1}$$

and all the partial derivatives $\partial y / \partial p_i$ can be calculated, either analytically or numerically. For each observed value y_{obs}, we calculate a value y_{calc} and minimize the value of a weighted residual wR:

$$wR = \sqrt{\frac{\sum_i w_i (y_{obs}(i) - y_{calc}(i))^2}{\sum_i w_i y_{obs}(i)^2}} \tag{2.2}$$

Here each difference is multiplied by a weight w_i that reflects the uncertainties of the experimental values. In case of crystal structure analysis, the observed values would be the observed intensities in a diffraction pattern and the calculated values those intensities that were calculated based on a structural model. Model parameters will be the lattice parameters, the positions of the atoms in the unit cell, atomic displacement parameters etc. as well as experimental parameters, such as the background. Under the assumption that we have a periodic crystal, the partial derivatives of the intensity with respect to lattice parameters, atom positions etc., can all be derived analytically.

For disordered structures, the situation becomes more complicated. Except for a few special cases like stacking faults or short-range order problems, no general analytical function straightforwardly links the disorder parameter to the intensity. The intensity can still be calculated from structural models. The simulation, however, usually involves the application of random choices to generate part or all of the atom positions, and the analytical derivative of the intensity with respect to the order parameter is no longer available. A numeric calculation of the derivatives involves the repeated simulation of a new model for each parameter and is very time consuming.
Other, general optimization problems also do not have an analytical derivative. For the traveling salesman problem, for example, no derivative exists between the sequence in which the spots are visited and the resulting length of the trip.
Under these circumstances, optimization algorithms are required that can find the best solution without calculation of the partial derivatives. Evolutionary algorithms are such an alternative to least squares refinement algorithms.
These algorithms loosely mimic the evolution of plants and animals under an environmental pressure. In contrast to a least squares algorithm they usually do not operate with a single parameter set but a group of parameter sets.

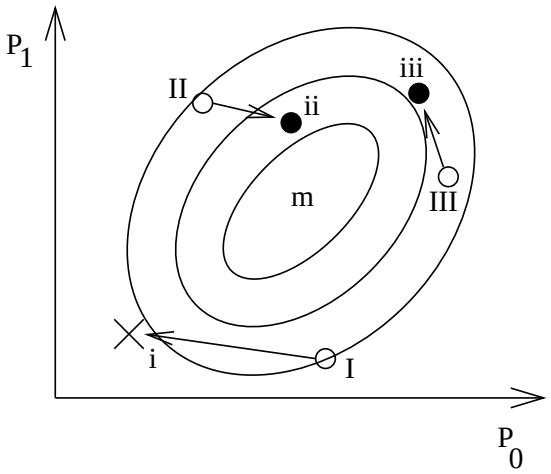


Figure 2.1: Schematic sketch of an evolutionary algorithm. Two parameters P_0 and P_1 are refined in the search for the optimum solution. The ellipsoidal lines represent lines of equal R-values. The global minimum is at point m. Members I, II, and III generate the new members i, ii, and iii by modification of the original parameter values. The new members ii and iii have an improved R-value, while member i has a worse R-value compared to the original member I.

The algorithm begins by creating a group of parameter sets, see Fig. 2.1. Each group member is a list of parameter values M:[$p_0, p_1, ..., p_n$], and for each member the R-value or more generally speaking the cost function is calculated. Within crystallographic applications, the R-value is usually calculated as described by Eq. 2.2
Next, a new group of parameter sets is generated. Several different algorithms for this step exist. One possibility is to change each parameter by a Gaussian distributed random number with mean zero, which is added to the original parameter value. The variance of this distribution will vary from parameter to parameter. The lattice constants, for example will need another

variance than the angles. In Fig. 2.1 the members I, II, and III create in turn the new members i, ii, and iii. This modification of a single parent member is comparable to the genetic mutation in biological systems. Most systems apply a second modification, that mixes the parameter values of two different parent members, a process that roughly resembles the sexual replication.
Once the new group of members has been generated, their R-values are calculated in turn. In the example child members ii, and iii have a smaller R-value compared to their respective parent members, while child i has a higher R-value than its parent. The next task to decide which members of the old parents and children shall be retained and form the next parent group from which the next children group is to be generated. A number of different approaches exist for this task. One option is to compare a child with its immediate parent and to keep the better of these two. In the situation depicted in Fig. 2.1, this would choose the parameter sets I, ii, and iii as parents for the next generation. Another approach is to combine all parents and all children into one group. From this group the N best are chosen as parents for the next generation. In Fig. 2.1 this would give the parameter sets ii, III, and iii as parents for the next generation, since these three sets have the three lowest R-values.
As the parameter values change from generation to generations, those members survive that have better R-values. As consequence, the parameter value evolve towards the minimum R-value. If the landscape of R-values is more rugged compared to the simple situation of Fig. 2.1, one has to be careful not to refine into a local minimum instead of the global minimum. The literature on evolutionary algorithms extensively deals with this issue.

2.2 The differential evolutionary algorithm

The essential feature of the differential evolutionary algorithm, introduced by Price and Storn Price et al. (2005), is the process, by which new children are generated.

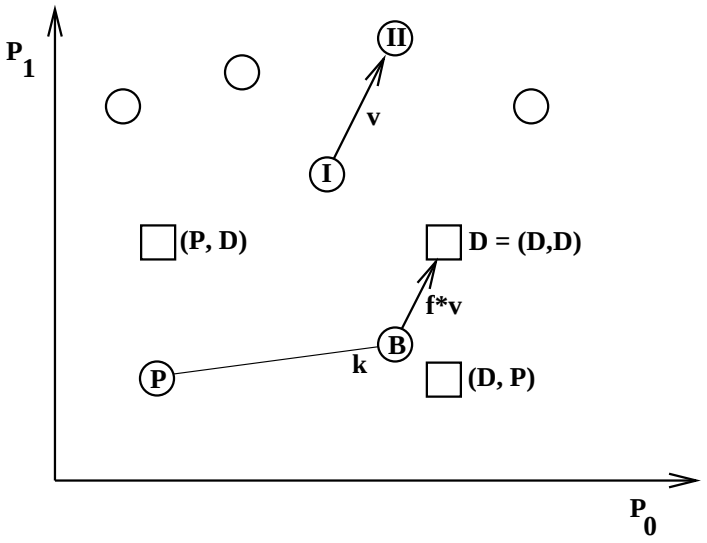


Figure 2.2: Schematic diagram of the differential evolution algorithm.

The algorithm picks in random sequence all members of the current generation, shown as circles in Fig. 2.2. The current parent has been marked by P in the figure. Another member, the

base B is chosen at random. Next, two other members are also chosen at random, members I and II in Fig. 2.2. The differential evolutionary algorithm then calculates the difference vector $\vec{v} = \vec{II} - \vec{I}$ between these two parents. This difference is the part of the algorithm that coined the name. The difference vector is multiplied by a factor f and added to the base member. All four members are different members of the population. The scale factor f is a variable that is used to control the refinement properties of the differential evolutionary algorithm. In general it should be somewhat smaller than one. The sum of the base member and the difference vector is a general point in parameter space, called the donor D. The donor D is the basic modification of th original parent vector. In contrast to other evolutionary algorithms, there is no direct connection between the parameter values of the parent P and its modification, the donor D.
After the determination of the donor D, the differential evolutionary algorithm allows for a mixing of the parameter values between the parent P and the donor D. By random choice, parameter values are taken either from the donor or from the parent. To ensure that the parent is not replicated, one parameter value is always taken from the donor D. The probability, by which the other parameters are taken from the donor D is called the cross over probability. Fig. 2.2 also illustrates the cross over process. If both parameters happen to be taken from the donor, the final child is the donor itself. If parameter p_0 is taken from the donor and parameter p_1 taken from the parent, the child will be the position labeled (D,P). Alternatively, if parameter p_0 is taken from the parent and parameter p_1 from the donor, the child will be the position labeled (P,D). Thus the cross over leads to a mixing of the parameter values of the donor D and the parent P. It depends on the refinement problem at hand, whether the cross over probability should favor parameters of the donor or the parent in order to ensure convergence.
Once children have been created for all parents, their R-values are computed. The differential evolutionary algorithm compares the R-values of each parent and its immediate child. Whoever has the lower R-value survives and is treated as parent for the next generation.
Several modifications to this basic differential evolutionary algorithm exist. The first modification concerns the choice of the donor base B. In the standard algorithm, the donor base is chosen randomly among the members of the population. As an alternative one can decide to take the current best member as donor base for all children. This will search predominantly in the neighborhood of the current best member and thus speed up the convergence into the minimum close to the current best member. If, however, this minimum is a local instead of the global minimum, chances are higher that all children will be within this local minimum as well.
Another alternative allows to add the scaled difference vector to any point along a straight line between parent and donor base, the line marked k in Fig. 2.2. A control variable *k* chooses the point. I the usual definition, the donor base is chosen if k=1 and the parent if k=0, and any point in between for intermediate values of k. In principle k is not limited to the interval [0:1], and DIFFEV does not limit your choice.
Choosing the surviving members allows for another modification of the original algorithm. Instead of a pairwise comparison of parent and child, one can also group all M parents and all N children into one group. Those M member of this combined group that have the lowest R-values survive and are used as parents for the next generation. The status of original parent or child is not taken into account. This approach will usually lead to a faster convergence, albeit at the risk of convergence into a local minimum. If the number of children is increased beyond the number of parents, the *evolutionary pressure* increases and the convergence is generally enhanced.

2.3 Termination criteria

Several different criteria may be used to terminate an evolutionary refinement.

- **Global minimum has been reached**
  If the values that correspond to the global minimum is known, the refinement can stop, once the lowest trial value falls within a defined threshold above this value. Unfortunately, in case of structure refinements, the lowest R-value cannot be known before hand and this criterion is not well suited.

- **Predefined number of refinement cycles**
  If the best R-value does not decrease for a given number of generations, chances are that we are very close to the global minimum. Unfortunately, one can never know whether the refinement may not improve after just a another few generations. This criterion may, however, be used to determine a good refinement strategy. The main variables to the differential evolutionary algorithm are the population size, the scale factor f by which the difference vector is multiplied, and the cross over probability. If refinements with different settings for these control parameters are allowed to run for a given number of generations. Those control parameters that lead to the lowest R-values after these generations can then be taken as good parameters for further refinement problems.

- **Population statistics**
  For diffraction data, it is straightforward to calculate an expected R-value. The refinement can be stopped, once the best R-value reaches this value, or at least comes close. Another choice could be to wait until all R-values have dropped to within a defined range of R-values above the expected R-value. The corresponding parameter range may then be inspected to determine the corresponding parameter uncertainties. If the model is insufficient, one may never reach this situation. Instead one could terminate the refinement once all R-values have become very similar to each other. If the lowest R-value is significantly above the R-expected one should run the refinement again with different starting parameters, of different control parameter settings to exclude convergence into a local minimum. If the parameters refine into the same minimum, the model should be analyzed and hopefully be improved.

- **User intervention** The last choice is to run the refinement indefinitely and to terminate the refinement manually be the user. Given the many different disorder problems that DIFFEV may face, this is the main termination criterion offered at present.

2.4 Optimizing the performance

The examples in this section use the example from chapter 3. Details given here are sketchy, refer to chapter 3 for full details.
Choosing the best setup for the refinement is in itself an abstract optimization task. One should choose those values for the control parameters that will cause the refinement to find the global minimum with the least amount of function calls. With regards to the differential evolutionary algorithm one needs to select the best values for:

- **Population size**
- **Scale factor f**
- **Cross over probability**
- **Choice of donor base**
- **Selection mode**
- **Local search probability**

The actual values that give the best performance will depend on the refinement problem at hand. See the discussion in chapters 2 and 3 of Price et al. (2005) for a further details. In the following we show a few examples that illustrate how to find good parameter settings.

2.4.1 Population size

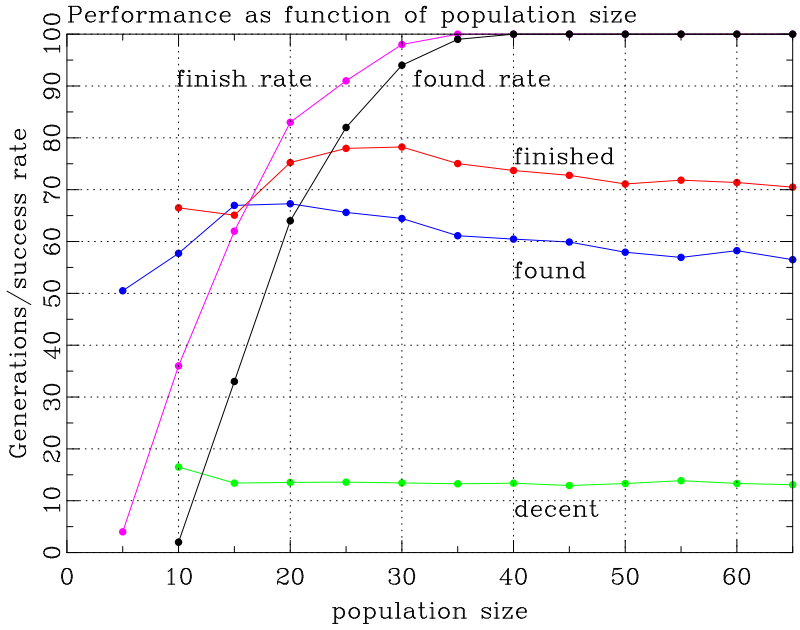


Figure 2.3: Success rate as function of population size.

If the population size is small, few permutations exist between the members of the population and there will be few locations that are searched in the parameter space.
Fig. 2.3 shows the effect the population size has on the refinement of the modified arctan function. This function requires three parameters, and parameters 2 and 3 pose a challenge due to the high noise present in the data. The true parameters for the example function were $P_1 = 100$; $P_2 = 100.23$; $P_3 = 0.1$. The refinement was run for population sizes from 5 to 65 members. The donor base was chosen at random, and the selection mode took the best members from the combined group of parents and children. The local search mode was switched off. The refinement was considered successful, if the R-value fell below a given threshold. Due

to the special function, this meant that the parameters are close to the true parameters and that the refinement will from here on converge into the global minimum. The refinement was allowed to run for 100 generations. Refinements that did not reach the global minimum or did not finish to refine to the global minimum were considered failures. At each population size, the refinement was repeated 100 times. Fig. 2.3 shows the performance as function of population size. The blue and red curves show the number of generations required to get close to the global minimum, respectively to finish refining into the global minimum. The black and purple curves show the percentage of refinements that came close to the minimum, respectively finished refining into the global minimum within the allowed 100 generations.
A population size of about 40 members is needed to ensure a 100% success rate. At smaller population sizes, a larger fraction of the refinements does not find the minimum and thus the refinement cannot be considered satisfactory. If the population size is increased beyond 40, the number of generations required to get close to the minimum, respectively to finish refining into the global minimum decreases slightly. The increase in function calls due to the population size increase is, however, higher than the gain obtained by faster refinements.

2.4.2 Scale factor and cross over probability

To test good values for these two parameters, the population size of 40 was adapted according to the observations on the population size. Both factors were chosen randomly in the interval [0:1]. As in the previous investigation, the refinement was considered successful, if the parameters reached the true values within 100 generations.

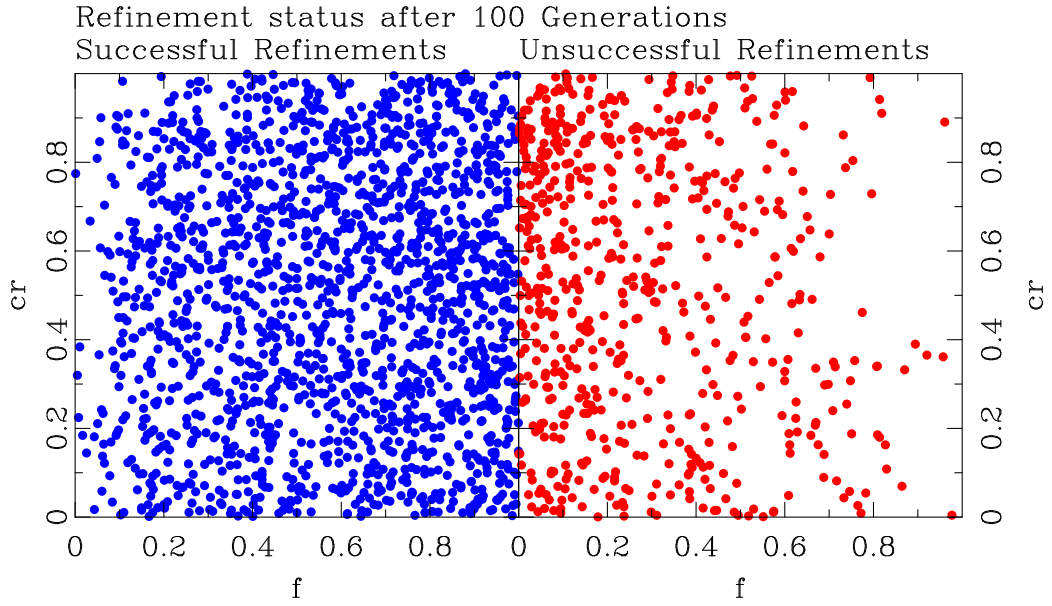


Figure 2.4: Success rate as function of scale factor f and cross over probability cr.

The figure shows that successful refinements of this example function require a scale factor that should be closer to 1. For very small scale factors, the algorithm effectively searches in the local environment of the donor base instead of a wide parameter range. In this example, this increases the chance of refining into a local minimum.

2.4.3 Local search probability

A cut through R-value space along P_3 at P_1 and P_2 at their respective optimum values shows a narrow minimum around the optimum value of 0.1. This behavior might indicate that a local search around the members might enhance the convergence into this narrow minimum. To test this, two different test were performed, in which the local search probability was systematically varied from 0 to 1.
For both tests with different local search options, otherwise identical control variables were used. The population size was set to 40 members, the scale factor was 0.81 and the cross over probability 0.8. The donor base was chosen randomly and the best members of the combined parent/children group were selected. For each local search probability the refinement was repeated 100 times.
Figure 2.5 shows the number of generations required to find a parameter combination close to the global minimum (blue), the total number of generations required to refine very close to the global minimum (red), the number of generations required from the time the first parameter set is close to the global minimum until the global minimum is reached (green), the percentage of refinements that came close to the minimum (black, and the percentage of refinements that finished refining into the global minimum (purple).

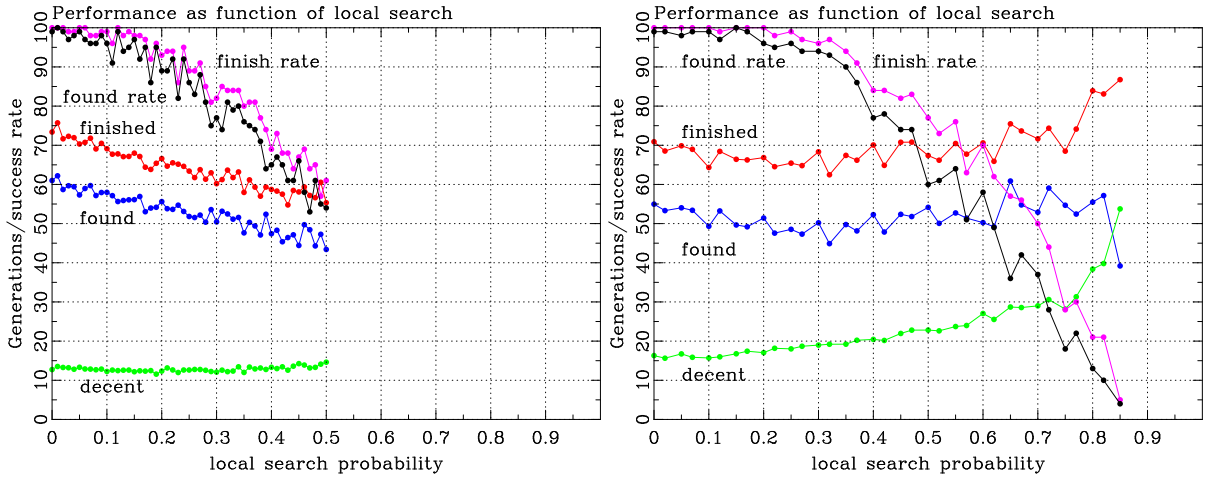


Figure 2.5: Refinement behavior as function of local search probability. Left, local search sigma adapted to 0.02 of the parameter spread, Right, local search sigma at fixed values (0.1; 0.1; 0.002).

The refinement with adaptable local search sigma showed that the number of generations required to find the global minimum decreases slightly with increasing local search probability. Beyond a probability of roughly 15%, the percentage of successful refinements starts to decline quickly. For the refinements with fixed local search sigma, the effect on the number of cycles is less pronounced. All in all one can see that the local search does not provide a clear effect on the refinement efficiency. In general, the original differential refinement algorithm seems to work better.

2.4.4 Limiting Parameters

Sometimes it is advantageous to fix one or several parameters in order to be able to perform a limited search. This is mostly helpful in the early stages of a refinement. Here one can refine a limited set of parameters just to get a rough idea where the likely global minimum is located. If parameters are not correlated one can limit the refinement to a small subset or even to a single parameter. The optimized value for this parameter should be reasonably close to the global minimum, if the parameter is that is refined is not correlated to the other parameters. The main advantage of such a search is that the dimension of the search space is much smaller and thus a much smaller population size will be sufficient.
To `fix` and to `unfix` a parameter, DIFFEV offers the two commands `fix` and `release` with the respective syntax:

```
fix P_lata, best
fix P_latx, 5.00
```

With the second parameter set to `best`, the parameter is set to the value of the member with the current lowest cost function, i.e. R-value. Alternatively you can set the parameter to another value, with the limitation that the parameter must be within the absolute boundaries for this parameter that were set with the `newpara` command or by assigning a value to the variables `pop_xmin[]` and `pop_xmax[]`.
For all subsequent cycles, the value of this parameter will be fixed to the chosen value. As this will change the parameter combination of all but the best member, the respective R-values for the next refinement cycle are invalidated to ensure correct testing of the next generation with respect to the generation prioor to the `fix` command.
The complementary `release` command allows you to set a new range of trial parameters for this parameter:

```
release P_lata, range:sigma
release P_latc, range:sigma, value:set_point, min:par_min, max:par_max
```

In the simplest form, the parameter is initialized again within a window of +-sigma around the value of the current best member.
The absolute window for this parameter is set to a range of +-3*sigma around the value of the current best member.
This new absoute window might be too wide or may include parameter values outside a physically sensible limit. As an example, a atomic displacement parameter may end up with a negative lower boundary. To prevent this behavior, the optional parameters `min:par_min` and `max:par_max` allow yo to fix the absolute boundaries to proper values.
The optional parameter `value:set_point` allows you to center the initialization window at a value `set_point` instead of the automatic centering around the current best value.

2.5 Invoking the slave program

The typical refinement requires the following steps:

- definition of the problem

- initialization
- A loop over the required generations within each loop the simulation of all crystal structures and the calculation of all cost functions / R-Values
- A comparison of old and new cost function values / R-values and generation of new trial parameters.

In the first step the number of population members and the number of refine-able parameters and their allowed range must be specified. DIFFEV furthermore expects the definition of log files to keep track of the refinement. See the examples in 3 for details on the commands.
The initialization step will assign starting values to all parameters that you want to refine. DIFFEV expects you to provide for each parameter a range within which the parameters are allowed, and a (narrower) range for the starting distribution. The initialization will place the starting parameter values with an even random distribution with the starting window. See 3 for further details.
As of version 5.3.0 the trial parameters need not be written to a file on the disk. If DIFFEV is used within the DISCUS SUITE, the trial parameters can be transferred directly to the slave program via the command: 'init silent'.
Within the loop the slave program that will simulate the crystal and evaluate the cost function / R-Value must be started. As of version 5.4.0 a unified command 'run_mpi' should be used for this purpose. The command will recognize whether DIFFEV runs as stand alone program or as part of the discus_suite (the strongly recommended style) and if the program uses MPI to process the slave program in parallel. An identical macro for the slave program can be used in all cases.
The slave program is actually started by DIFFEV as:

```
discus -macro discus_main.mac PWD kid indiv > discus_log.xxxx.yyyy
```

If DIFFEV is part of the DISCUS SUITE, the suite will internally switch to the discus section to execute the macro 'discus_main.mac'. If DIFFEV is run as a stand alone program (highly discouraged), it uses a system call to start the DISCUS program with the command line option to execute the macro. In both cases identical DISCUS macros can be used. The one exception is the 'silent' option for the 'init' and 'compare' commands. This option is available within the DISCUS SUITE only. For the stand alone version of DIFFEV no direct communication exists between DIFFEV and DISCUS. The trial parameters and the R-values need to be written to disk to communicate between the programs. It is strongly recommended to use the DISCUS SUITE.

2.5.1 Parallel Refinement via evolutionary algorithms

As the refinement in DIFFEV is based on a large population, it naturally lends itself to parallel performance. This version of DIFFEV has a build in support for a MPI based parallel distribution. Within this model, each of the children is simulated / calculated in parallel to each other. On a large scale computing facility you will typically submit your job to a queue and you will have to request a specific number of nodes and over all wall time. As example we will use the queue system at the high performance center in Erlangen. Take this as a general guide and refer to your system for changes that you might have to do.

Within this general set up the simulations of all crystal structures can be performed independently and thus in parallel. If the calculation of the r-value takes considerable time this can be performed in parallel as well. This parallel calculation must only be carried out once the simulation is finished.
If the simulation involves small crystal structures, or a distribution of defects and/or sizes, you might need to simulate several individual structures for each member of the population. Again these simulations can be performed in parallel.
MPI is a widely distributed system to run jobs in parallel. You can even use it effectively on a multi core computer. To use DIFFEV with MPI you need to turn on the MPI option during compilation or run the MPI version from the DISCUS download server.
A set of command to run the refinement in parallel will be like:

```
set prompt, redirect
@diffev_setup.mac       ! all the set up commands
init silent             ! Initialize the population
do i[0]=1,10            ! Make a loop over 10 refinement cycles
   run_mpi discus, discus_main.mac, repeat:20, compute:parallel, logfile:discus_log
   run_mpi kuplot, kuplot_main.mac, repeat:20, compute:serial,   logfile:kuplot_log
   compare
enddo
exit
```

The `run_mpi` command instructs DIFFEV to run the program specified as first parameter in parallel. The second parameter is the macro that the slave program will execute.
Starting with version 5.17.0 the remaining parameters take on the form for optional commands. Backward compatibility holds.
The parameter `repeat:` tells the slave section how often a calculation for an member has to be repeated. Such a repetition is often necessary if you simulate small crystals with defects that are distributed with any kind of randomness. In these cases an individual crystal is not a good representative of the many possible defect distributions. You will get a reliable result only if the simulation is repeated and the corresponding data /diffraction /PDF/...) are averaged.
The parameter `compute:` tells the slave section whether it should calculate these individual repetitions serially within a single macro or whether the individual simulations will be calculated in parallel alongside the parallel distribution of all the members.
The parameter `logfile:` enables you to save the output from the slave section into a log file. This will be helpful during the development of your refinement. Once everything works fine, omit this parameter or set its value to `none` or to`/dev/null` to discard the output. The log file parameter will be augmented by a four digit number with leading zeros for each member. If the individual repetitions are done in parallel, the filename is augmented by a further four digit number that stands for the current individual repetition.
The slave section is started by DIFFEV as:

```
discus -macro discus_main.mac > discus_log.xxxx.yyyy
```

The refinement parameters, the child number and the individual repetition number are transferred internally and can be accessed with the macro `discus_main.mac`. The log file name is extended by two four digit numbers xxxx and yyyy, which hold the values of the child and the individual repetition.
DIFFEV uses several variables with fixed name that are transferred to the slave section. These are:

- REF_GENERATION The current refinement generation number
- REF_MEMBER The population size
- REF_CHILDREN The number of children for which the simulation is carried out.
- REF_DIMENSION The number of parameters to be refined
- REF_KID The current child number
- REF_INDIV The individual repetition number.
- REF_NINDIV The total number of individual repetitions that are needed.
- INDI_PARALLEL Is the calculation of the individual repetitions carried out serially by the macro `discus_main.mac` or distributed in parallel.

Within DIFFEV the parameters to be refined are set by the command `newpara`. The name that you give to the parameter on this command is stored as a user defined variable and given its proper current value. Use this variable name within the macro `discus_main.mac` as well. See the second example in3 for further details.
The number of parallel instances that will run depends on your MPI system. On a stand alone computer with several cores you will start DIFFEV as

```
mpiexec -n 8 discus_suite -macro refine_main.mac
```

For the DISCUS SUITE , the main refinement macro should switch to the DIFFEV section to the refinement and return to the DISCUS SUITE as in:

```
diffev
... setup
... init silent
... loop over generations
exit   ! This goes back to the SUITE
exit   ! This will finish the SUITE
```

In this example, MPI will reserve 8 cores for your job, execute DIFFEV as a section of the DISCUS SUITE , which must be located in a directory where the standard PATH environment variable will find it. The macro `refine_main.mac` must contain all instructions for DIFFEV, including the `set prompt, redirect` and final `exit` commands.
The MPI scheduler does not know how long a calculation by the slave program may take. This might cause an idle state for some or almost all CPU's once almost all children in a given generation have finished. This will cause your system not to run as efficiently as possible. Some of this idle state cannot be avoided. You can reduce the idle state if you keep the number of requested CPU's much smaller than the number of simulations required for one generation. Under these settings, each CPU will have to run several simulations and you can hope that the work load will average out. To average the calculations DIFFEV sends the CHILDREN times NINDIV calculations in a double loop. The inner, faster index is over all CHILDREN, the slower over all individual calculations.
As of version 5.17.0 the DIFFEV section has been optimized, see the section 2.5.2 in this chapter.

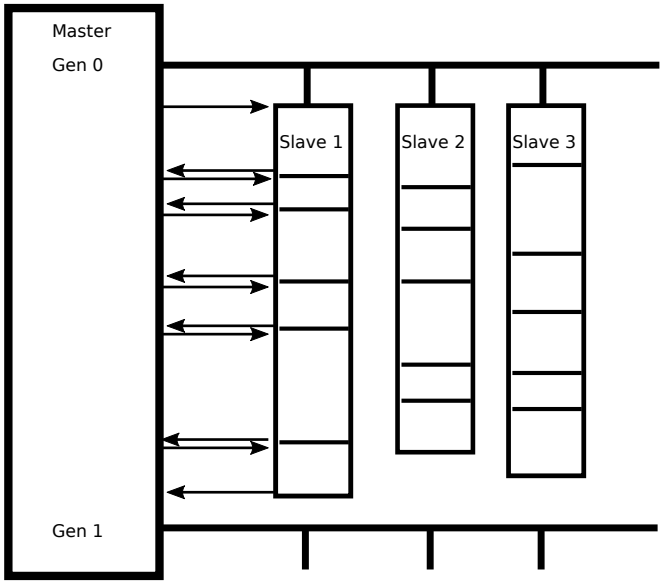


Figure 2.6: Overview of parallel refinement process

2.5.2 Computational aspects of parallel refinement

In this section we will cover some computational aspects of the refinement that may guide you to optimize the performance. Fig 2.6 shows a schematic overview of the parallel refinement. When you start the DISCUS SUITE via mpi as in

```
mpiexec -np 8 diffev -macro refinement.mac
```

the DISCUS SUITE starts a master process that will in turn start np-1 slave processes. In each generation the master process will instruct the slaves to perform several tasks. Lets say you you have a large population and each individual simulation does not have to be repeated. The master will then instruct the slaves to start the calculation for the first np-1 members. As soon as a slave is finished, it will send back the result to the master and receive the instructions to start the calculations for the next member. This communication is indicated by the arrows between the master and the first slave in Fig 2.6. In many cases the calculation times for each member will differ as indicated by the different vertical positions of the horizontal bars within each slave. Ideally all slaves would terminate their last calculation at the same time. This would minimize idle time on the computer. This is, however, difficult to enforce and some idle time for almost all slave can hardly be avoided.

If the computation time for all members is identical, the number of slaves and the population size should be matched such that the population size is an integer multiple of the number of slaves np-1. This will ensure that all slaves will perform calculations for an identical number of members and will be finished at the same time. If the population size is larger by one member, a single slave will be busy with this last member while all other slaves are just waiting for new instructions.

If the computation times for the different members differ but can be estimated ahead of times it is best to start to distribute the large jobs first and then once these are done to hand out

the shorter jobs. Those slaves that receive the shorter jobs will perform calculations for more members that those slaves that receive the larger jobs. If the large jobs are handed out first, the chances are optimized that all the shorter jobs will finish roughly at the same time.
If the computation times cannot be estimated ahead of times, some idle time is likely to occur. To reduce the amount of idle times each slave should perform calculations for several members. As in the last paragraph, those slaves that happen to receive larger jobs will perform calculations for fewer members. With several jobs per slave the fluctuations at the end should be about half the computation time for an individual average job. If, as the other extreme, each slave were to calculate just one simulation, the one slave that takes the longest calculation will force all other slaves to be idle until it is finished. This will cause a huge amount of idle time.
At the moment the DISCUS SUITE does not attempt to estimate how long each calculation will take. There are just too many different parameters that will influence the time like nanoparticle size, number of atoms, number of atom types, time spend within a Monte-Carlo simulation etc. Given this lack of information the idle time is minimized if each slave performs several calculations.
The situation is slightly different if the calculations require the need to perform each simulation several times for a given member. This will be the case for example if you simulate small nanoparticles and the simulation involves the creation of (randomly distributed) defects. As the individual nanoparticle is small, its structure and as a consequence its powder diffraction pattern or PDF will not be a good representative of all possible structural conformations. You will have to repeat the simulation again for the same set of parameters. Afterwards all the individual powder pattern /PDF's will have to be averaged. If an individual member has to be repeated N times, the total number of simulations will of course be N times as large. To reduce the idle time the DISCUS SUITE will hand out jobs for the first individual calculation for each member first. Once all these are done the next individual calculations will follow. All individual repetitions for an individual member likely require a similar amount of time, while the requirements for different members may vary.
Once all members and all individual repetitions have been performed the DISCUS SUITE will have to average the temporary results. This averaging process can of course be performed in parallel as well. This raises two further issues related to the architecture of the computer at hand.
A local computer or local compute server has an architecture similar to Fig. 2.7. Several CPUs or cores perform the computing and MPI can distribute the workload onto these CPUs or cores. The CPUs share a common disk to store permanent data. At a large scale compute facility many of these units are combined. The local storage at each of these nodes often is a solid state memory instead of a traditional hard disk. A very large central disk provides storage space for global results. The scheduling process at such a high performance compute center (HPC) typically give your job a privileged and sole access to all CPUs at a node and you process may request several nodes at the same time. The communication between the CPU's on a node and their local disk is fast, efficient and only interferes with you own jobs. The communication to the central hard disk is common to all processes by all users at the HPC. Thus this communication should be reduced as much as possible as it may slow down your own process and that of other users as well.
The MPI distribution usually does not offer a direct assignment of a particular job to a specific core. This means, that the master process usually has no means to predict on which CPU a

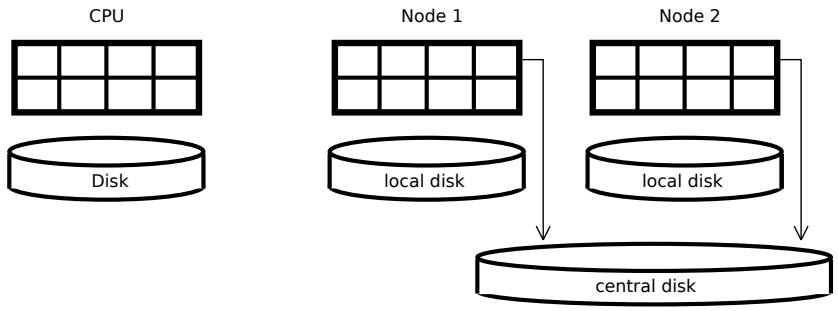


Figure 2.7: Simplified architecture of single PC with eight core CPU, left, and multi node HPC, right

given calculation will occur.
As of version 5.17.0 and later ones, the DIFFEV section tries to optimize the internal distribution of jobs onto the available nodes and their individual cores as best as possible for any compute environment.
Keep in mind that we can have two different refinement situations:

- The calculation for a member does not have to be repeated several times
- The calculation for a member does have to be repeated several times and the results need to be averaged once all calculations have been performed

Independent of this you may have two different compute architectures available for your refinement:

- A local PC or local small scale server with a single node
- A HPC center with many nodes, each node has its own local storages common to all cores on this node

To match your refinement situation to the local architecture and to optimize the performance DIFFEV offers two different refinement styles:

```
variable integer, nindiv
REF_NINDIV = 20
... diffev setup ...
pop_n[0] = 192
pop_c[0] = 192
... further diffev setup ...
... FIRST style
run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:/dev/null
...
... SECOND style
run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:parallel, logfile:/dev/null
run_mpi kuplot, kup.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:/dev/null
```

Within the first style DIFFEV will start pop_c calculations. here in this example arbitrarily set to 192. These 192 calculations will be distributed in parallel onto the available CPUs. If the model requires any repetitions, the DISCUS macro here `dis.diffev.mac` will have to include a loop over these repetitions. The DISCUS macro also needs to include a `branch` to the KUPLOT section to calculate the R-value. As all repetitions for a given member are performed serially on

a single CPU, you can write the DISCUS output data either onto a hard disk or directly into the KUPLOT memory space. The latter is done by prepending the output file name by the string `kuplot`. See the DISCUS manual for further details. As the individual calculations during the refinement are temporary data of no long term relevance, it is a good idea to avoid disk input/output. Thus the internal write is recommended.
Within the second style DIFFEV will start pop_c times nindiv calculations. In this example this results in a total of 3840 calculations that can all be performed in parallel. Each time the DISCUS macro `dis.diffev.mac` will calculate data for just one member and will have to write these data onto a disk. Once all 3820 calculations are finished, DIFFEV will instruct KUPLOT in a second parallel job to average all calculations for a given member and to calculate the corresponding R-value. As a single KUPLOT slave needs to average all repetitions, the parameter `"compute"` is set to `"serial"`. In this second model you need to write the data onto a disk, as the memory requirements would be too large for internal storage. A more stringent reason to write the data onto disk is that a single slave will perform calculations for members and individual repetions in an unpredictable sequence. One can expect that the calculations for one member are performed by different slaves.
Now lets have a look at strategies to run these two different schemes on the two compute architectures. Common to both architectures, you will have np-1 processors available to perform actual calculations while the one master processor keeps itself busy with administrative tasks. All slave processors should perform several jobs as this minimizes the risk of a long idle time at the end of a generation.

- Single PC or small server
  On these systems several users might be active or you may run several refinements or other tasks. As these tasks can take up idle time the following rules can be relaxed.
  - No individual repetitions. The number of members should be an integer multiple of (np-1). To reduce the idle time the multiple should be high.
  - With repetitions. Both computation models are an option. If the calculation times are long per repetition you will not do too many disk I/Os to slow you down or to risk the integrity of your hardware. If the calculation time is fairly short, it might be kinder to your hardware to choose th first computational model as temporary data can (and should) be saved internally.
- High Performance Compute center
  On these systems you will have sole access to a given node and its CPU. Local disk I/O is fast and does not cause wear and tear if the storage is a solid state disk. As you are the sole user, idle time should be minimized. A lot of disk I/O across the network onto the central disk will be frowned upon and should be reduced as much as possible.
  - No individual repetitions. The number of members should be an integer multiple of (np-1). To reduce the idle time the multiple should be high. As a single node often has 24 or more CPUs request few nodes and long wall time instead.
  - With repetitions. Both computation models are an option. The second model makes better use of the many processors available. As of version 5.17.0 DIFFEV will place all individual repetitions for one member onto the same node. If the local storage at

a node is a solid state disk the temporary storage will not significantly slow down your process. Thus different slave processes that run on the same node can calculate individual repetitions for any of the members on that node. The number of members should still be an integer multiple of (np-1) but the factor can be reduced. The (large) number of individual repetitions will level out the workload at each core of a given node.

Chapter 3

Example refinements

The example in this chapter will be used to explain in detail the commands and control variables that DIFFEV uses. The example is a simple function y = F(x). While this restriction eases the display of the data, and refinement process, it does not impose a restriction on the settings. The macros and input data needed to run this example are found along with the DIFFEV source code, in the directory *diffev/TESTS*.

3.1 Nanoparticle refinement

In this section a full blown refinement of a disordered nanoparticle structure is illustrated, which is also to be found in the DISCUS book, Neder and Proffen (2007). The data for the nanoparticle are taken from Neder et al. (2007). The nanoparticle sample consists of ZnSe nanoparticles. The PDF was derived from X-ray powder diffraction data taken at beam line BW5 at the former storage ring DORIS at DESY, Germany.
The experimental PDF suggests nanoparticles with diameter around 3 to 4 nm. As bulk ZnSe crystallizes in the Zincblende structure, this provides for a good starting model. Further modifications include an ellipsoidal shape rather than a spherical shape and the inclusion of stacking faults that allow for an alternation of locally cubic and hexagonal sequences. The first interatomic distances are observed at 2.44Å and at 3.98Å only. These two distances impose a locally tetrahedral environment, and the stacking faults have to preserve this environment. TEM images indicated that lattice fringes across the particles appeared undisturbed. This is consistent with a model in which one of the symmetrically equivalent cubic [111] directions is along the long axis of the ellipsoids. Stacking faults will shift the layers normal to this direction and apparently no stacking faults are present along any of the other [111] directions.
As the nanoparticles are small, an individual simulated particle that contains stacking faults will not be a good representation of all possible configurations. See Neder and Proffen (2007) for a full discussion. In the refinement we will thus have to calculate individual nanoparticles several times for each set of parameters and average the corresponding calculated PDFs.
The main refinement macro consists of the following steps, which are shortly outline before we go into the full details:

```
diffev                        ! Switch to diffev section
#
  @cleanup.mac                ! Remove old files
```

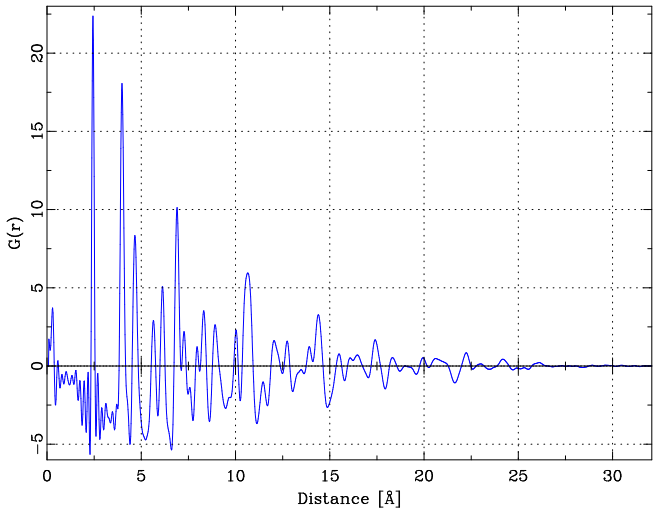


Figure 3.1: Experimental PDF for ZnSe nanoparticles

```
#
  REF_NINDIV = 20               ! Define number of individual repetitions
  @get_model.mac                ! Define the refinement model to use
  @diffev_setup.mac             ! Define refinement details
#
  init silent                   ! initialize all parameters, silently no files are written
#                               ! The command can be omitted
#
  do i[0]=1, 200                ! Number of refinement cycles
    echo "Loop GENERATION %4d %4d", i[0], REF_GENERATION
    !                           ! Run discus section with macro dis.diffev.mac, no repetitions
    run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:LOGFILES/d
    compare silent              ! Compare R-values to previous generation
  enddo
exit                            ! Back to suite
exit                            ! Finish the suite as well
```

With the first line the DISCUS SUITE is instructed to switch to the DIFFEV section. The `cleanup.mac` macro contains line to remove all files from a previous refinement run. The commands work on all operating systems and do not ask to confirm the removal of the files. Next the line REF_NINDIV defines the number of individual repetitions that need to be performed for each parameter set i.e. for each member of the population.
If you look at the remainder of the main macro, you'll notice that the vairable REF_NINDIV corresponds to the parameter for the optional `repeat:` on the `run_mpi` line. You can run the individual repetitions either serially in `dis.diffev.mac` with `compute:serial` or run parallel versions of this macro. These models are explained further down after this quick overview. For this example, the user can specify four different nanoparticle types:

- a spherical particle with no stacking faults
- a spherical particle with stacking faults
- an ellipsoidal particle with no stacking faults
- an ellipsoidal particle with stacking faults

Macro `get_model.mac` reads the users choice. If the user had chosen a particle with no stacking faults, there will be no need to average several individual particles. Accordingly, the choice in the main refinement macro is overwritten by setting the value of `REF_NINDIV` to one. Next, macro `diffev_setup.mac` does all the detailed setup for the refinement. Once the setup is done, `init silent` initializes the first generation. As of version 5.16 this command can actually be omitted, as DIFFEV recognizes the state it is in and will perform an initialization if needed.
In the following loop over a fixed number of refinement cycles DIFFEV uses the DISCUS section to simulate the nanoparticles and to calculate the agreement between the experimental and calculated PDFs. The `compare` command instructs DIFFEV to compare the current generation to it parents and to create a next generation of trial parameters. The final two lines finish the DIFFEV section and the DISCUS SUITE program itself.
On a Unix type operating system the DISCUS SUITE program will be started with a line like:

```
mpiexec -np 250 discus_suite -macro refine.mac
```

Alternatively you can usually use `mpirun` and most MPI environments will take either `np` (the proper form) or `n` to specify the number of parallel processes to be started.
With Windows start a first DISCUS SUITE window and at the command prompt type:

```
parallel refine.mac
```

An optional parameter allows control over the number of parallel processes as well:

```
parallel 250, refine.mac
```

The number of parallel processes that you want to start depends on the hardware that is available. On a single PC, do not choose more processes than the number of cores available. The additional administrative burden will actually slow down the over all performance. On a high performance compute facility consult the local manual. Usually you will will submit a job into a queue. On a modern system you will have a total of several hundred processors that are available.
From the model of an ellipsoidal nanoparticle with stacking faults we can derive 7 structural parameters. Although bulk ZnSe crystallizes in the cubic Zincblende structure, we will choose to describe the nanoparticle based on the hexagonal Wurtzite structure. The cubic lattice parameters can be transferred into the hexagonal metric via the matrix:

$$(\vec{a}_h, \vec{b}_h, \vec{c}_h) = \left(\vec{a}_c, \vec{b}_c, \vec{c}_c\right) \begin{pmatrix} 1/2 & 0 & 1 \\ -1/2 & 1/2 & 1 \\ 0 & -1/2 & 1 \end{pmatrix} \tag{3.1}$$

As the Wurtzite structure consists of two alternating layers, while the Zincblende structure consists of three layers, the Wurtzite lattice parameters are actually better obtained by:

$$(\vec{a}_h, \vec{b}_h, \vec{c}_h) = \left(\vec{a}_c, \vec{b}_c, \vec{c}_c\right) \begin{pmatrix} 1/2 & 0 & 1/3 \\ -1/2 & 1/2 & 1/3 \\ 0 & -1/2 & 1/3 \end{pmatrix} \tag{3.2}$$

This transformation yields hexagonal lattice parameters a=4.008Å c=6.5444Å. One can expect slightly smaller values for the current nanoparticles, as the measurement was carried out at 30K. The second neighbor distance at 3.98 is a good estimate for the hexagonal a lattice parameter. In an ideal Wurtzite structure both atom types are located at Wyckoff position 2a 3/3, 1/3, z with z=0 and z=3/8 and we will use this value as starting value for the refinement. As Zn (30) and Se (34) are very close to each other in the table of elements, and the data were collected with X-rays, it is not important which element is placed at z=0. Furthermore, as both sites are symmetrically equivalent, it is a good approximation to use just one common atomic displacement parameter for both elements. The experimental PDF yields significant distances up to approximately 30Å, and we will use this as initial estimate for the nanoparticle diameter. This gives us seven structural parameters:

- lattice parameter a
- lattice parameter c
- z-position of Zn
- a common isotropic displacement parameter
- stacking fault parameter
- diameter in the hexagonal a-b plane
- diameter along the hexagonal c-axis

Beyond this we have parameters for the PDF:

- number density
- the linear correlation term
- the quadratic correlation term
- the instrumental dampening term
- the instrumental broadening term
- the value of Q_{max}
- the scale factor

The instrumental terms and Q_{max} are of course fixed during the refinement. For this refinement the quadratic correlation term is also fixed at zero, and the linear correlation term is refined. We will calculate the PDF from an actual simulation of individual nanoparticles that will contain randomly placed stacking faults. As a consequence, the $-4\pi\rho_0 r$ line needs to be corrected for the finite size and shape of the nanoparticle. Currently DISCUS does not contain any analytical corrections other than that for a spherical shape. Instead the parameters for an empirical shape function will be refined together with the structural model. As this empirical shape function is

completely correlated to the number density we will fix the number density at zero and refine the scale factor only.
In total this gives nine parameters to refine. For this number of parameters a population size of 90 to 140 members is a good compromise.
With these initial considerations we are ready for a detailed look at the refinement macros. The first macro to consider in detail is `diffev_setup.mac`. The essential lines in the macro are:

```
 1: pop_gen[1] =   0
 2: pop_n[1]   =  90
 3: pop_c[1]   =  90
 4: newpara    P_lata, range:[3.9000,  4.020],  value:[3.900, 4.020], status:free
 5: diff_cr[1] = 0.9
 6: diff_f[1]  = 0.81
 7: #
 8: donor      random
 9: selection  best,all
10: #
11: #
12: lastfile   DIFFEV/Current
13: logfile    DIFFEV/Parameter
14: summary    DIFFEV/Summary
15: backup TEMP/calc, FINAL/final
```

The first line sets the generation number to zero. Although this is the default value at program start, this command is necessary to ensure that DISCUS will properly start a new refinement. The next two lines set the population and children size to 90, ten times the number of parameters to be refined. In the authors experience it is best to keep the number of parents (= the population size) and the number of children created in each generation at equal values.
Next we define a (first) new parameter, called `P_lata`. You are free to choose any parameter name as long as it conforms to the variable name restrictions within the DISCUS SUITE, and consists of 1 to 16 characters. As of version 5.16 this parameter name is placed into the list of user defined variables and its value is passed down to the slave process. Thus, in this example, we will use the variable `P_lata` whenever we need to reference the a lattice parameter within DISCUS. The first optional parameter `range:[]` sets two numerical parameters which are fixed lower and upper limits for the parameter value. This bracket the expected value of 3.98Å. As this window is fairly small already, the starting window is set to identical values as well, parameter `value:[]` . DISCUS will assume that this parameter is a real valued parameter rather than an integer valued one. As we want to refine this parameter, its status is set to `status:fixed`.
The essential parameter is `value:[]`, it is not optional. DIFFEV attempts to guess sensible values for the `range:[]` if absent. If `status` is absent, DIFFEV assumes a refined parameter if the values are not identical.
Further lines have to added for each of the parameters that we wish to refine. You will find these in the actual macro.
Lines 5 and six set the behavior of the cross-over and the scale factor parameters for the differential evolutionary algorithm. In contrast to many other numerical tasks Price et al. (2005), the exact values of the cross-over probability and the scale factor do not seem to be very relevant for the refinement of diffraction data. Keep these parameters as set.
The DIFFEV section offers two choices to determine the donor for each parent. You can take the donor as the current best member in hope that all new children will be close to this member and

thus hopefully will yield a low R-value. The alternative choice is to take a randomly chosen member as donor and to add the difference vector to this member. This latter strategy seems to work better and is chosen in this example.
For the selection of the members that will produce the following generation, line 9, two choices are available. The original algorithm makes a choice between a parent and its child. The better of these two survives to act as parent for the following generation, even if both the parent and the child have a rather high R-value compared to the total population. It is a bit more efficient to pool all parents and all children into a common group and to select the better half of this group as parents for the next generation, as is done in this example in line 9.
Lines 12 to 14 define the files into which DIFFEV will write a log of the refinement. As the DISCUS SUITE can pass the trial parameters directly to the slave process and receive back the R-value the need for the trial and result files has become obsolete. By setting these to a value `silent` their use is switched off. The other three lines define the full log for the currently finished generation, the full log for the complete refinement and the short summary file for the full refinement. If, as in this example, the files are placed into a folder, DIFFEV will not check if this folder does exist. You need to create the folder prior to the refinement.
In the last line the backup option is defined. DIFFEV will expect that the slave processes write a diffraction pattern/PDF into the folder `TEMP` with filenames `calc.0001`, where the four digit number corresponds to the child number. The currently best files are archived into the folder `FINAL` as files `final.0001` etc. This archiving is helpful if you want to monitor the progress of a lengthy refinement.
The next step in the main macro is the command `init silent`. This command instructs DIFFEV to assign initial values to all parameters. The values are randomly chosen in the start interval that is defined by the last two parameters on the `newpara` command. The `silent` parameter indicates the no trialfiles are to be written. Instead the DISCUS SUITE will pass the parameters internally to the slave processes. As of version 5.6 this command is not needed. The DIFFEV section will initialize all parameters in generation zero even without this command.
This gets us back to the main refinement macro. It performs a loop over 200 or so refinement cycles. DIFFEV does allow the user to stop a refinement upon a variety of convergence criteria. As population based refinements tend to be stagnant for a while, and since high performance centers usually allot a fixed time per run, it is best to choose a fixed number of cycles. The main workload in each generation is distributed via the command

```
run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:LOGFILES/d
```

This line instructs DIFFEVto switch to the DISCUS section and to execute (in parallel) the macro `dis.diffev.mac`. The third parameter tells the DIFFEV section how often an individual calculation is to be repeated on the DIFFEV side. With `compute:serial` we tell DIFFEV and the DISCUS slaves that the individual repetitions shall be calculated serially within the macro `dis.diffev.mac`.
The last parameter is the base of logfiles that will receive all output from DISCUS . The base is augmented by a four digit number for each member of the population. In the example above files `LOGFILES/d.0001` etc would be written into the folder `LOGFILES` whose existence you need to ensure prior to the refinement. Once you are happy with the DISCUS macros the parameter can be omitted or be replaced by `none` or `/dev/null`. In this case the output is discarded.

If the repetitions are performed on the DIFFEV side, the number of tasks that need to be distributed and can be calculated in parallel is the population size times the number of individual repetitions. Once this is finished the individual results (powder diffraction pattern or PDFs) that belong to a given member need to be collected, averaged and the agreement to the experimental data be calculated. The combination of commands to do so would for example be:

```
nindiv   =  50
pop_c[1] = 100
run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:parallel, logfile:LOGFILES/d
run_mpi discus, kup.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:LOGFILES/d
```

Here we use `compute:parallel` to tell DIFFEV to run the DISCUS macro `dis.diffev.mac` `pop_c[0]*nindiv` times. The DISCUS macro should be written such that it performs exactly one calculation for a given member and individual repetition. The second parallel command `run_mpi` instructs KUPLOT to collect the nindiv results for a given member and to return the agreement factor (R-value) to the DIFFEV section. The maximum number of tasks that can run in parallel is the population size, her in this example 100. KUPLOT needs to know the number of individual repetitions, this is done by specifying the `repeat:REF_NINDIV` parameter as well. Since KUPLOT has to average all these individual calculated PDF data, it will work serially on all the individual repetitions thus the need for `compute:serial`.
The advantage of this computational model is that we can run a large number of tasks in parallel with the first `run_mpi` command. The disadvantage is the need to store, collect and average the individual diffraction pattern/PDFs on a central disk to which all the parallel tasks have to have access to.
The alternative model runs only the members of the population in a parallel fashion. The individual repetitions are performed one after the other on the one CPU that is calculating a given member. The two `run_mpi` lines above reduce to the one line stated originally:

```
run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:LOGFILES/d
```

As of version 5.17.0 the number of individual repetitions will still be handed down to `dis.diffev.mac`. The macro has to branch into KUPLOT in order to average individual calculations. The disadvantage of this model is the much smaller number of tasks that can be run in parallel. The number of parallel tasks should only be a fraction of the population size. This will ensure that each node or CPU will perform several calculations and all nodes will -hopefully- finish at the approximately same time. The advantage of this model is the considerably reduced data I/O. All individual results do not have to be written onto the hard disk at all. Instead DISCUS can write these directly into the KUPLOT memory.
Whether to perform individual repetitions on the DIFFEV side or on the DISCUS side depends on the computational hardware that is available. On a local PC you typically have one node with likely some four to eight CPUs, one common storage disk, and possibly a smaller yet faster (temporary) solid state disk. At a HPC you usually have many nodes that work in parallel each with several CPUs. While there is one common storage disk, each nodes will have its own (temporary) and fast storage as well. The central storage is optimized towards huge files that are written every once in a while. Small temporary file input and output is to be avoided onto this central storage, as it put a huge burden on the communication between the nodes.
Unfortunately the message passing interface (MPI) that DIFFEV uses to perform the refinement in parallel does not allow an easy assignment of individual tasks onto a specific node, nor

does it have an easy way to let the central master program know on which node a given task was calculated. This limitation favors the model where individual repetitions are performed serially by the DISCUS macro. Future hybrid releases of the DISCUS SUITE may improve at this point. As of version 5.17.0 DIFFEV will place all individual repetitions of a given member onto the same node, eliminating the MPI disadvantages. Thus a compute model with parallel distribution of the individual repetitions is recommended.
The essential lines in the macro `dis.diffev.mac` are:

```
 1: set error,exit
 2: variable integer, indiv
 3: @get_model.mac
 4: @global.mac
 5: #
 6: do indiv = 1, REF_NINDIV
 7:   @discus.znse.mac
 8: enddo
 9: branch   kuplot    !Switch to KUPLOT
10:    @kup.diffev.mac
11: exit
```

This is a fairly generic macro that needs little change from one refinement to another. Line 1 sets a strict termination, if any error should occur in subsequent macros. This prevents a refinement from running on and on if something went wrong in the refinement. Line 2 defines a local variable for the loop in lines 6 to 8. In this loop we go over all the individual repetitions for the current parameter set, i.e. for the current member of the population.
Line 3 might have to be adapted to a different refinement. In this examples it determines the model that is to be used. The macro is identical to the one used in the main refinement macro.
Line 4 runs a macro `global.mac` that is used to set several filenames. By collecting all these file name definitions in one central macro it becomes less tedious to adapt to different substances, data files etc.
Lines 6 to 8 run a loop over all the individual repetitions that are required if a simulation involves randomly placed defects. Often an individual object created in a simulation is not large enough to hold a good representation of the defect distribution. In these cases one has to build several structures according to the same rules. In a second step all the diffraction pattern or PDFs need then to be averaged. If the model does not involve statistically distributed defects, all that you have to do is omit the loop or more simply set the number of iterations `nindiv` to one. The macro `get_model.mac` does exactly this, if the user decides on a model of a perfect crystal without stacking faults. For this model, all nanoparticles are identical and it is sufficient to create just a single one.
In line 7 we run the main work horse `discus.znse.mac`. This macro has of course to be adapted in detail for a different type of refinement. We will go through this macro in the following section.
In lines 9 and 10 we branch from the DISCUS section to the KUPLOT section in order to average the calculated PDFs and to calculate the agreement factor to the observed data.
Finally, in line 11 the DISCUS section is terminated which will return the control to the master process. If there are still tasks to be distributed the current slave process will receive a new task and repeat the steps discussed for this macro.
The style adopted in this macro `dis.diffev.mac` corresponds to the first style defined in section 2.5.1. The individual repetitions are computed serially by one slave process. Accord-

ingly we will use internal storage of the individual output data and calculate the R-value in the branch to KUPLOT in lines 9 and 10. For this style a suitable `global.mac` would be:

```
 1: variable character, TMPDIR    ! temporary DISCUS files   'internal'     or '.'
 2: variable character, INDIDIR   ! temporary PDF directory  'kuplot'       or '.'
 3: variable character, DATADIR   ! input data directory     '/tmp/mpkr04/' or '.'
 4: variable character, DATAFILE  ! Input Data File
 5: variable character, SUBSTANCE ! Substance name
 6: DATADIR   = "%c",'/tmp/zzzzzz'
 7: DATAFILE  = "%c",'PD32ZS56.DAT'
 8: INDIDIR   = "%c",'kuplot'
 9: TMPDIR    = "%c",'internal'
10: SUBSTANCE = "%c",'znse_wurtzite'
```

The KUPLOT macro `kup.diffev.mac` will copy the experimental data `DATAFILE` to a temporary location at `DATADIR`. This serves to reduce network traffic during the refinement, see the macro `kup.diffev.mac` further down. Make sure that the directory `/tmp/zzzzz` exists and that you have write privileges. At a local PC or local small server you could use the line

```
 6: DATADIR   = "%c",'.'
```

instead. This would place the copies to a local directory in your current working directory. `TMPDIR` is a directory to temporary DISCUS structure files. Except for debugging purposes you can keep this as `internal`, which will place these files into the internal memory structure of DISCUS without unnecessary disk input/output. As this compute style will calculate all individual repetitions serially in one call to macro `dis.diffev.mac`, we do not have to write these files to a disk. See the macro `pdf.mac` further down to see how the string `INDIDIR` is placed at the beginning of these temporary output files. As the string has the value `kuplot` DISCUS is instructed not to write the files onto disk but to copy them directly into the KUPLOT memory.
For the second style in 2.5.1 we leave it to DIFFEV to distribute the calculation of individual repetitions to separate slave processes. in this case macro `dis.diffev.mac` should read:

```
 1: set error,exit
 2: variable integer, indiv
 3: @get_model.mac
 4: @global.mac
 5: #
 6: indiv = REF_INDIV
 7:   @discus.znse.mac
 8: enddo
 9: exit
```

Note that in line 6 we assign the value of `REF_INDIV` to our local variable `indiv`. `REF_INDIV` is a global variable set by DIFFEV along with `REF_KID` to the individual repetition number. This assignment allows us to use all further macros without change. The branch to KUPLOT is omitted, as the main refinement macro `refine.mac` will now include a separate `run_mpi` statement to run the KUPLOT calculations.
The only change needed for `global.mac` is:

```
 9: INDIDIR   = "%c",'/tmp/zzzzzz'
```

As the DISCUS SUITE (starting with version 5.17.0) places all individual repetitions onto the same node of your cluster, these temporary output files for one member of the population will

all be at the same local directory `/tmp/zzzzzz` and thus be accessible to the second `run_mpi` step in which KUPLOT is used and the Rvalue is calculated.
The main simulation macro `discus.znse.mac` consists of the essential lines:

```
 1: variable real, P_qmax
 2: P_qmax      = 30.00
 3: #
 4: read
 5:    free P_lata, P_lata, P_latc, 90.00, 90.00, 120.00, P63mc
 6: insert Zn, 1./3., 2./3., P_z_zn, P_biso
 7: insert Se, 1./3., 2./3., 0.0000, P_biso
 8: #
 9: if(model.eq.PERFECT_SPHERE .or. model.eq.PERFECT_ELLIPSOID) then
 0:    P_stack  = 1.00
11: endif
12: #
13: if(model.eq.PERFECT_SPHERE .or. model.eq.STACKED_SPHERE) then
14:    P_cc_dia     = P_ab_dia  !ref_para[ 06]    ! Spherical model
15: endif
16: #
17: save
18:   outfile "%c/STRU/%c.%4D.%4D.cell", TMPDIR,SUBSTANCE,REF_KID,indiv
19:   run
20: exit
21: @makelayers.mac
22: #
23: @shape.ellipsoid.mac
24: @pdf.mac
```

In lines 1 and 2 we begin by defining the Q_{max} value that was used in the actual experiment to derive the experimental PDF. It would be sufficient to write this number explicitly in the pdf macro that is executed in line 24. I prefer to write this value at this point, as this makes the pdf macro a completely generic macro that does not have to be changed at all for the refinement of another nanoparticle pdf. If the number id encoded in the actual macro it increases the chance to forget to change this number.
Lines 4 to 7 are used to create the asymmetric unit for the Wurtzite type ZnSe structure. Check the DISCUS manual for full details of these commands. The important point to mention here is that all variables in this macro that start with `P_` are parameters that are to be refined. The parameters were defined in the macro `diffev_setup.mac`. OK, the exception to this rule is `P_qmax` the fixed value for Q_max. DIFFEV does not impose any rule on the parameter names that you choose, as long as they are regular variables that are valid in any section of the DISCUS SUITE and whose name consist of up to 16 characters. The `free` command creates general coordinates system, here a hexagonal coordinates system. The last parameter defines the space group intended. The Wurtzite structure consists of two atoms in space group $P6_3mc$. Both atoms are at the position 1/3, 2/3, z. We can fix the z position of one of the two atoms at an arbitrary value. Here Se is fixed to z=0, while the z position of Zn is to be refined. Its z value will be close to 3/8, the value in the idealized Wurtzite structure.
In lines 9 to 11 and 13 to 15, respectively, restrictions that are imposed by the model are applied to the parameters. If the user intends to model a perfect nanoparticle without stacking faults, the parameter is set to one, which will result in a perfect Zincblende structure type. Similarly, if a spherical nanoparticle is intended, the diameter `P_cc_dia` along the hexagonal c-axis is fixed to a value identical to the diameter in the a-b plane. These local changes to the parameter values do not affect the value of the parameters within the scope of the master process. If

the master process tries to refine these parameters, no sensible results will be obtained for the parameters if the local restrictions are applied.
As lines 4 to 7 have created the asymmetric unit only, this asymmetric unit is saved in lines 17 to 20. This will enable later parts of the macro to read the modified asymmetric unit and to expand it to a full crystal. The value of the character variables `TMPDIR` and `SUBSTANCE` is set within macro `global.mac`. For this example the variable `TMPDIR` is set to a value `internal`. DISCUS saves any file that starts with the string "internal" within an internal memory structure rather than to write the file to the hard disk. As the current asymmetric unit is a temporary file there is no need to write the file. During the development of a refinement it might be a good idea to write the file to the hard disk in order to ease debugging processes. As the refinement shall eventually run in parallel, it is mandatory that all files that are written to the hard disk have a unique file name. For that reason the current member number and individual repetition number are added to the file name.
In this refinement we intend to stack layers of the hexagonal Wurtzite structure type in order to create a disordered sequence of layers. As the interatomic distances are to be refined, the layer structures will vary as function of the refinement parameters. Macro `makelayers.mac` in line 22 creates these new layers that are needed for the Wurtzite/Zincblende stacking sequence.
The macro `shape.ellipsoid.mac` does the actual stacking process and includes commands to shape the particle into an ellipsoidal shape.
Finally in line 24 macro `pdf.mac` is used to calculate the PDF for this current simulation.
Once the PDF is calculated the macro `dis.diffev.mac` continues with the average process and evaluation of the R-value through the KUPLOT section.
The discussion will continue with a detailed look at the nanoparticle build up. First lets look at `makelayers.mac`
The atom positions in the (idealized) Wurtzite type structure are located at:

```
Zn  2/3,  1/3, -1/8     A
Se  1/3,  2/3,  0       b
Zn  1/3,  2/3,  3/8     B
Se  2/3,  1/3,  1/2     a
Zn  2/3,  1/3,  7/8     A
Se  1/3,  2/3,  1       b
...
```

The letters ABAB denote the layer sequence of the Zn atoms, the small letters abab the corresponding sequence of the Se atoms. The whole structure can be described as a sequence of (Ab)(Ba)(Ab)(Ba) double layers. Within a (Ab) double layer the vector from Zn to Se is [-1/3, +1/3, 1/8], while the corresponding vector in the (Ba) double layer is [+1/3, -1/3, 1/8]. The Wurtzite structure can thus be described as a ABAB Sequence of two different layer types (Ab) and (Ba). The (Ba) double layer can be created by rotating the (Ab) double layer by 180Â° around the c-axis. With this notation, the perfect Zincblende structure type results if either pure (Ab) double layers or pure (Ba) double layers are stacked in a sequence of ABCABC... Mixed layer types result if the sequence of (Ab) and (Ba) layers is modified.

```
1: # makelayers.mac
2: #
3: variable integer,width
4: variable real, thick
5: width = int(2.5 * P_ab_dia/P_lata )
6: #
```

```
 7: read
 8:    cell "%c/STRU/%c.%4D.%4D.cell",TMPDIR,SUBSTANCE,REF_KID,indiv,width,width,2
 9: thick = blen(0.0, 0.0, 0.5-P_z_zn) + 0.1
10: #
11: surface
12:    boundary hkl, 0, 0, 1,   0.5,inside
13:    boundary hkl, 0, 0,-1, thick,inside
14: exit
15: purge
16: #
17: save
18:    outf "%c/STRU/%c.%4D.%4D.layer",TMPDIR,SUBSTANCE,REF_KID,indiv
19:    write all
20:    run
21: exit
22: #
23: symm
24:    angle 180.0
25:    type  proper
26:    mode  repl
27:    sel   all
28:    incl  all
29:    orig  0.333333, 0.666667, 0.00
30:    uvw   0,0,1
31:    trans 0.00, 0.00, 0.00
32:    run
33: exit
34: #
35: save
36:    outf "%c/STRU/%c.%4D.%4D.rotated",TMPDIR,SUBSTANCE,REF_KID,indiv
37:    run
38: exit
```

Macro `makelayers.mac` builds these two layer types. Two strategies could be used to perform this task. You could either build a large template of (Ab) and (Ba) and then modify the atom coordinates of Zn relative to Se to adjust the refined lattice parameters and the z-position of Zn. Alternatively the whole layer can be recreated from the modified asymmetric unit. Here the latter strategy es employed. The allows for more flexibility to build a layer of arbitrary size to ensure that the final nanoparticle will fit inside the layer.
In line 5 variable width is calculated to be large enough to include the final nanoparticle with diameter `P_ab_dia` in the hexagonal a-b plane. The factor 2.5 is a bit generous to ensure that at no corner of the hexagonal space we will cut off atoms inside the intended diameter. In lines 7 and 8 the asymmetric template generated in `discus.znse.mac` is read and expanded to a width by width by 2 unit cells. The `surface` menu in lines 11 to 14 cuts off all atoms above the (0,0,1) surface located at a distance of 0.5 Å above the origin and all atoms below the (0,0,-1) surface located ad a distance of `thick` below the origin. The value of `thick` is calculated from the refinement parameter `P_z_zn` as idealized 1/2 - 3/8 = 1/8 along the c-axis. This places the (0,0,-1) plane just below the position of the Zn atoms at 2/3, 1/3, -1/8. All other atoms are no longer needed and removed by the `purge` prior to saving the layer in lines 17 to 21. Next the layer is rotated by 180° around the c-axis. Note that the rotation axis is placed at the Se position 1/3, 2/3, 0 to ensure correct stacking later on.
After these preparations we are now ready to build the actual nanoparticle inside our next macro `shape_ellipsoid.mac`.

```
 1: read
 2:    cell "%c/STRU/%c.%4D.%4D.cell",TMPDIR,SUBSTANCE,REF_KID,indiv
```

```
 3: #
 4: @stack.mac
 5: #
 6: surface
 7:    boundary ellipsoid, P_ab_dia, P_ab_dia, P_cc_dia, centx:0.0,centy:0.0,centz:P_cc_dia*0.5/P_latc
 8: exit
 9: purge
```

The macro is short and all of its lines could easily have been placed inside the main macro `discus.znse.mac` without making the main macro too long and too difficult to follow. Subdividing the macros does make it easier to modify the main macro by just changing one line.
The macro begins by reading the template again and expanding it to just one unit cell. This is done to ensure that the stacking sequence in `stack.mac` works on the correct metric, even if the main macro might later on be changed to work on further structures. Macro `stack.mac` does build the actual particle. Once it is done the `surface` menu is used to shape the crystal into a triaxial ellipsoid with diameters P_ab_dia, P_ab_dia, and P_cc_dia. Even though we are in hexagonal space, the three half axes of the ellipsoid are orthogonal to each other. See the DISCUS manual and on-line help for further details. The center of the ellipsoid is placed at half the diameter along the c-axis. The `stack` menu in DISCUS always places the first layer at the origin, thus the need to shift the center of the ellipsoid.
The essential lines in `stack.mac` are:

```
 1: variable integer,height
 2: height = int(2.*P_cc_dia/lat[3] + 2)
 3: #
 4: stack
 5:    layer "%c/STRU/%c.%4D.%4D.layer"  ,TMPDIR,SUBSTANCE,REF_KID,indiv
 6:    layer "%c/STRU/%c.%4D.%4D.rotated",TMPDIR,SUBSTANCE,REF_KID,indiv
 7:    trans  1,1, -0.3333, 0.3333, 0.5000
 8:    trans  1,2,  0.3333,-0.3333, 0.5000
 9:    trans  2,1, -0.3333, 0.3333, 0.5000
10:    trans  2,2,  0.3333,-0.3333, 0.5000
11: #
12:    number height
13: #
14:    distr  matrix
25:    crow   1,       P_stack, 1.00-P_stack
26:    crow   2,  1.00-P_stack,      P_stack
27: #
18:    aver   0.00, 0.00, 1.00
19:    modu   1.00, 0.00, 0.00,  0.00, 1.00, 0.00
20:    set mod ,on
21:    set trans,fixed
22: #
23:   create
24:   run
25: exit
```

We start by calculating the height of the stack in numbers of layers, line 2 and line 12. The factor "2" in the calculation is due to the fact that there are two of the double layers per unit cell in the Wurtzite structure type. Two further layers are added to avoid cutting off atoms due to rounding errors if the ellipsoid diameter is almost an integer multiple of the c lattice parameter. Within the `stack` menu (line 4) we first define the two layer types that comprise the stacking sequence (lines 5,6). The four `trans` commands define the translation vector from one layer type to the next layer type. Lines 8 and 9 are the vectors within a perfect Wurtzite structure

type, a perfect Zincblende structure type results with a sequence of layer types 1 only (line 7) or layer types 2 only (line 10).
DISCUS offers two different modes to determine the actual stacking sequence. In this example we use a probability matrix (line 14). The matrix elements define the probability that a layer type is followed by another layer type. As we have two layer types, we need the two rows of this matrix (lines 25, 26). Element (1,1) = P_stack gives the probability that a layer of type 1 is followed by a layer of type 1. Likewise element (1,2) = 1 - P_stack gives the probability that a layer of type 1 is followed by a layer of type 2. In this example, a value of P_stack close to +1 would result in an (almost) perfect Zincblende type sequence of layers of type 1, all shifted by the translation vector (line 7) [-1/3, 1/3, 1/2]. If the first layer happens to be chosen by the program to be type 2 an analogous Zincblende sequence of layer type 2 would result. A value of just about zero on the other hand, would result in an (almost) perfect Wurtzite type alternation of layer types 1 and 2, separated by vectors (line 8) [1/3, -1/3, 1/2] for type 1 followed by type 2 and (line 9) [-1/3, 1/3, 1/2] for layer type 2 followed by layer type 1.
In order to ease the final shaping of the ellipsoid, lines 18 to 21 define an average growth direction along vector [0,0,1] (line 18). As soon as the translation vectors from lines 7 to 10 place the next layer at a location further away from the c-axis than either of the two modulo vectors on line 19 the layer is shifted back by this modulo vector. As these two vectors [1,0,0] and [0,1,0] are integer lattice vectors this shift does not have any consequences other than to ensure that each layer origin is approximately at x and y equal to zero. Line 20 turns this modulo behavior on and line 21 ensures that the average growth direction is taken as the vector on line 18. Without this modulo operation a stacking fault parameter of 1 (or close to) would result in a shift of [-1/3, 1/3, 1/2] between any layer pair. The stack would be an oblique tower. For such an object the shaping of an ellipsoid becomes more tedious, as the required horizontal extend of each layer becomes linked to the vertical height and the stacking parameter.
Finally we `create` all layer origins (line 23) and build the actual list of atom coordinates within each layer (line 24). The separation of these two steps allows DISCUS a tremendous gain in speed when calculating the diffraction pattern for large crystals. Here, for the ellipsoidally shaped object both steps have to be done.
Finally macro `pdf.mac` calculates the pair distribution function (PDF) for the nanoparticles.

```
 1: @setup_pdf.mac
 2: #
 3: pdf
 4: #
 5:     ides all
 6:     jdes all
 7:     isel all
 8:     jsel all
 9: #
10:     set bound,    crystal, exact
11:     set dens,     P_density
12:     set corrlin,  P_corrlinear
13:     set corrquad, P_corrquad
14:     set srat,     P_srat,3.5
15:     set therm,    gauss
16:     set qbroad,   P_qbroad
17:     set qdamp,    P_qdamp
18:     set qmax,     P_qmax
19:     set rad,      xray
20:     set range,    pdf_range+5.0, 0.01
21:     set weig,     P_scale
```

```
22:     set finite,   periodic
23:     calc
24:     save pdf,"%c/INDI/indi.%4D.%4D"  ,INDIDIR, REF_KID, indiv
25: exit
```

In line 1 the variable `pdf_range` and its value are defined. This variable is used in line 20 to define the distance range over which the PDF is to be calculated. As the computation time scales with this value you might want to adapt it to test models over different refinement ranges or for differently sized nanoparticles. By placing this information into a short macro by itself, other parts of the macro suite can use the identical macro `setup_pdf.mac` and there is only one place that needs modification. After a clean selection of atom pairs for which to calculate the PDF (lines 5 to 8) all PDF parameters are set (lines 10 to 22). The PDF is calculated in line 23 and the data saved in line 24.
At this point macro `discus.znse.mac` returns control to macro `dis.diffev.mac` which in turn branches off to the KUPLOT section and performs the macro `kup.diffev.mac`

```
 1: variable integer,indiv       ! Local variable for repetition
 2: @global.mac                  ! Directory names
 3: @get_model.mac               ! Get the refinement model to use
 4: @kup.average.mac             ! Merge all individual calculations REF_NINDIV + 1
 5: load xy, "%c/DATA/%c.%4D", DATADIR, DATAFILE, REF_KID !              REF_NINDIV + 2
 6: spline REF_NINDIV+1, REF_NINDIV +2   ! Ensure identical x-axis-scale REF_NINDIV + 3
 7: @kup.fit.polynomial.mac      ! Correct 4PI RHO line and scale    REF_NINDIV + 7
 8: rval REF_NINDIV+2, REF_NINDIV+7, dat ! calc R-value, is transferred internally
 9: reset                        ! No DATA
10: exit                         ! Back to DISCUS / DIFFEV (depends on use)
```

In line 1 we define a variable `integer`, which is used inside the macro `kup.average.mac` to loop over the individual files. Lines 2 and 3 repeat the definitions encountered in the main macro `refine.mac`. The next macro `kup.average.mac` serves to average the individual files created in the main discus macro `discus.znse.mac`. Next, in line 5, we load the experimental PDF into KUPLOT . Line 6 could be omitted, if you make sure that the experimental data and the calculated data are on an identical x-scale. As it stands, the `spline` command tells KUPLOT to create a new data set, with the X-scale taken from the data set indicated by the second parameter. The y-scale is calculated as a spline function through the y-values of the data set specified by the first parameter. As the shape of the nanoparticles might be highly irregular, the PDF in `pdf.mac` was calculated with a number density of zero. The actual value and the corrections to the $-4\pi\rho_0 r$ line are applied in macro `kup.fit.polynomial.mac` (line 7). At this point we are ready to calculate the agreement between the experimental and calculated PDF in line 8. The last parameter signals that the experimental uncertainties are taken into account for the R-value calculation. A the `reset` in line 9 tells KUPLOT to remove all loaded data sets. Finally we `exit` the KUPLOT macro in line 10.
Note that in lines 6 and eight we do not refer to absolute data set numbers. Instead the data sets are labeled with reference to the number of individual repetitions `REF_NINDIV`. This is necessary, as the number of individual repetitions may change from refinement to refinement. The current macro should work for any number of repetitions. Before we look in detail at this numbering scheme, lets look into `kup.average.mac`

```
1: if(n[1]==0) then
2:   reset
3:   do indiv=1,REF_NINDIV
```

```
4:      load xy,"%c/INDI/indi.%4D.%4D", INDIDIR, REF_KID, indiv
5:   enddo
6: endif
7: merge all    ! creates new data set number: REF_NINDIV + 1
```

As explained above, the DISCUS macro may either save the calculated data sets onto a hard disk, or save them directly into KUPLOT . In the first case, KUPLOT will at this point be void of data sets i.e. $n[1] = 0$, and the `if` block will be executed. Within the block, the `REF_NINDIV` data sets are loaded in the loop. If DISCUS did save the calculations directly into KUPLOT , the `REF_NINDIV` data sets will already be loaded into KUPLOT . In both cases KUPLOT will at this point have loaded `REF_NINDIV` data sets. As this number may vary, is is best to refer to this relative number rather than to specify an absolute value in all subsequent KUPLOT macros. Finally, in line 7 these data sets are merged. As a consequence we will now have `REF_NINDIV + 1` data sets loaded into KUPLOT .
Loading the experimental PDF in line 5 of macro `kup.diffev.mac` places this experimental PDF into data set `REF_NINDIV+2`
Macro `kup.fit.polynomial.mac` fits a polynomial background to yield an empirical description of the $-4\pi\rho_0 r$ line.

```
 1: kcal sub,REF_NINDIV+2,REF_NINDIV+3 ! Creates data set REF_NINDIV + 4
 2: skal                       ! Fit needs to know x and y range
 3: fit REF_NINDIV + 4         ! Fit a function to data set REF_NINDIV + 4
 4:   func, poly,6             ! Polynomial of order x**6
 5:   para 1,0, 0.00           ! x^0 is fixed to 0.00
 6:   para 2,1, 1.00           ! x^1 refined, starts at 1.0
 7:   para 3,1, 0.00           ! number, flag, starting_value
 8:   para 4,1, 0.00
 9:   para 5,1, 0.00
10:   para 6,1, 0.00
11:   para 7,1, 0.00           ! x^6 refined, starts at 0.00
12:   wic  dat                 ! Use weights inherited from data set
13:   cycle 200                ! 200 cycles should be enough
14:   urf 0.5                  ! control refinement speed ~1/urf
15:   run                      ! Lets do it
16: exit                       ! back to main KUPLOT menu
17: kcal add,REF_NINDIV+3,REF_NINDIV + 5 ! Add polynomial to merged PDF's
18: ksav REF_NINDIV + 7           ! Save this calculated PDF
19:    outf "TEMP/calc.%4D",REF_KID
20:    run                     ! automatically returns to main menu
```

In line 1 we subtract the experimental PDF, data set `REF_NINDIV+3` from the merged calculated PDFs, data set `REF_NINDIV + 2`. This command stores the result in the next available data set `REF_NINDIV + 4`. Next, we scale the plot (although nothing is plotted to screen) as the fit needs to know the x-range over which we want to refine something. Within the `fit` menu (line) 3 we fit a calculated function to data set `REF_NINDIV + 4`. The function is defined in line 4 as a polynomial of order 6. This order ensures that quite complex nanoparticle shapes can be modelled. A (much) higher order would mask deficiencies of the nanoparticle model. Lines 5 to 11 define starting values for the seven parameters from $P_1 \times x^0$ to $P_7 \times x^6$. The three parameters to the `para` command are: Parameter number, refinement flag (0=fixed, 1=refined) and starting value. Starting values are 0.00 for all parameters except the linear term (line 6). The constant term (line 5) is fixed at zero to ensure that the polynomial goes through the origin at x=0.0, G(r) = 0.0.

The weighting scheme (line 12) uses the uncertainties inherited into the difference between experimental PDF and merged calculated PDF. Line 13 defines 200 refinement cycles, sufficient for this linear least squares problem. The `urf` parameter defines the speed by which KUPLOT changes its refined parameters from cycle to cycle. A large number means slower convergence as the parameters are changed by smaller amounts.
The `fit` menu creates two further data sets, the calculated function and the difference between the input data set and the calculated function. As the input data set is `REF_NINDIV + 4` (line 3) these new data sets are `REF_NINDIV + 5` and `REF_NINDIV + 6`. We now add (line 17) the calculated polynomial function (data set `REF_NINDIV + 5`) to the input data set. The resulting data set `REF_NINDIV + 7` is the final calculated PDF that will hopefully match the experimental PDF very well.
Lines 18 to 20 are used to save a copy of the final data set to hard disk. This save enables the use of the `backup` command in macro `diffev_setup.mac`. This option enables you to have a look at the current best calculated PDFs while the refinement is still ongoing. If you do not need this option, for example while you run a refinement on a high performance compute center you can omit these lines.
Control is returned to `kup.diffev.mac` which finishes with the calculation of the R-value (line 8), a reset (line 9) and the exit back to DISCUS .
KUPLOT calculates the aggreement between the two data sets according to Eq. 3.3.

$$wR = \sqrt{\frac{\sum_i w_i (y_{obs}(i) - y_{calc}(i))^2}{\sum_i w_i y_{obs}(i)^2}} \tag{3.3}$$

Here y_{obs} and y_{calc} are the observed and calculated function values, in our current example the PDF. The weighting scheme w_i assigns a weight to each individual data point. For a diffraction pattern the weight is commonly $1/\sigma^2$, where σ is the uncertainty of the experimental data point. Currently the commonly available data treatment software will not provide an uncertainty for each data point in the experimental PDF. A unit weight is most often used. As the errors in the PDF tend to decrease with increasing distance r, a weighting scheme with $\sigma \propto 1/r$ is also used. As the values of the PDF below a distance of approximately 1Åare not reliable, these data points should receive a low weight, respectivaly a high uncertainty.
At this point we have gone through all macros that are part of the nanoparticle refinement.

3.2 Noisy example function

The function in Fig. 3.2 shall be the data set for the refinement in this section. The function from which these data were calculated is:

$$y \;=\; P_1 atan\left(\frac{|x - P_2|}{P_3}\right) \tag{3.4}$$

where atan is the arc tangens or inverse tangens function, and P_i are free parameters that determine the behavior of the function. The function has a minimum y=0 at $x \;=\; P_2$. The width of this minimum depends on P_3. For small values of P_3 the minimum is smaller. Far away from the minimum, the function is almost constant at $y \;=\; \pi/2P_1$. The data were generated by adding a Gaussian distributed noise to each data point.

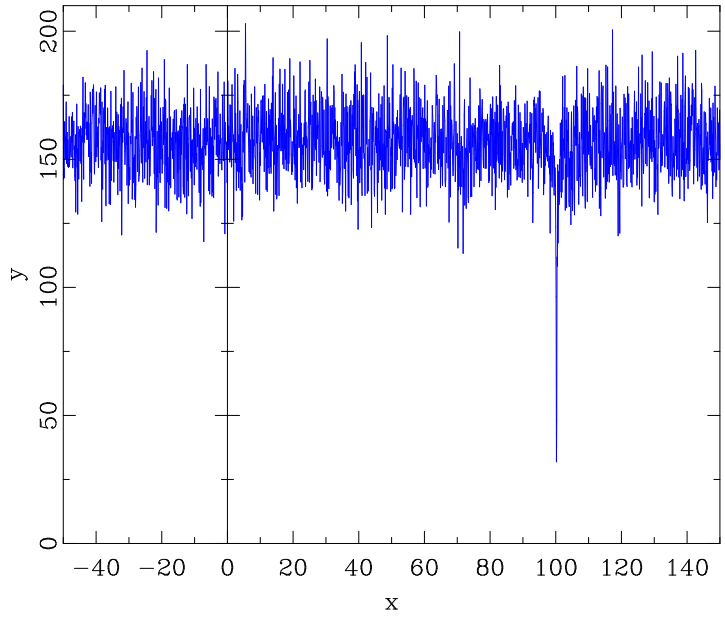


Figure 3.2: Example data

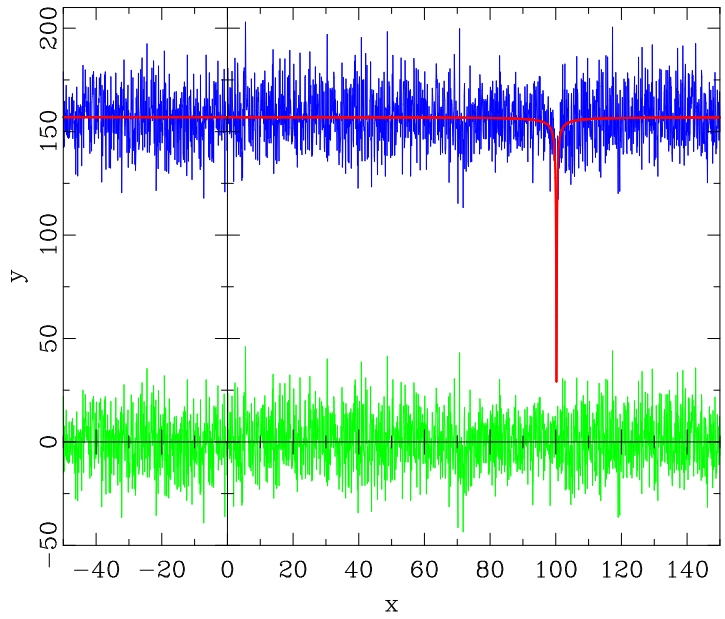


Figure 3.3: Example data in comparison to the ideal function. The R-value between the observed and calculated function is 8.18%.

This data set proves to be a very hard challenge for any least squares algorithm. Unless the starting values are very close to the actual values, the least squares algorithm will not find the global minimum. The very high noise level effectively prevents the algorithm from finding the position of the minimum. Instead the R-value is minimized by setting P_3 to the smallest positive value. At this level, the width of the function is reduced to one point, if any. Furthermore, the slope of the function is essentially zero for all x, and the position of the minimum does not influence the R-value. The R-value is, however, significantly above the R-value that is achieved for the ideal parameter values, show in Fig. 3.3.
To refine the function with DIFFEV, we need to define a couple of items. These include:

- Population size
- Limits, starting range etc. for the parameters P_i
- Values for refinement control variables
- location and names for files

The optimum size of a population is not always straightforward to estimate. Since the minimum at P_1 is very narrow, a fairly large population size is advisable. A test shows that the population should consist of approximately 40 members to ensure that the minimum is found. The corresponding commands would be:

```
pop_n[1]   = 40
pop_c[1]   = 40
pop_gen[1] =  0
```

Here the number of members and children was set to the same value, there is no fixed need to do so, unless the selection mode is set to compare a child with its immediate parent. With the last statement the current generation number is set to zero. If a refinement is to be continued, the generation number can be set to the corresponding value.
The next group of definitions includes the number of parameters, and then for each parameter a suitable name, hard boundaries, a starting range and definitions how to handle parameter behavior at the hard boundary and in a local search.
In the current example, we have three parameters, all of which are floating numbers. None of the parameters has any absolute lower or upper boundary imposed by external rules. Parameter P_3 must not be equal to zero, this we will handle later as a constraint. Sensible hard limits for the parameters could be:

parameter	lower boundary	upper boundary	comment
1	0	200	All data points are larger than zero.
2	-50	150	Minimum must be somewhere within the observed x-range
3	-1	1	Include zero to use constraints

If one has good estimates for the parameter values, these can be used to limit the initial spread of the population. A narrow spread of the initial population around the expected final value will speed up the convergence. Should these estimates prove to be wrong, however, the refinement will take extra long or may fail altogether. Here we will not impose any prior knowledge on any of the parameters and use the full range set by the hard boundaries.

During the initial refinement stages, the differences between the members will be large and chances are that the donor falls outside the allowed hard boundary interval. DIFFEV corrects this situation by setting the violating parameter to a Gaussian distributed value with mean at the hard boundary. The respective half of the Gaussian distribution that falls inside the boundary range is taken as valid region. The user can set the sigma for this distribution and define whether this sigma shall remain constant or be adapted during the refinement. For parameters 1 and 2 we will set the initial sigma to 1, and for parameter 3 to 0.02. In later refinement cycles the value of sigma can be adapted to a fraction of the total parameter spread, in our example to 0.2 of the parameter spread.
Similarly, the sigma for a local search is fixed to starting values and adapted to a fraction of 0.01 of the total parameter spread.
For parameter 1 this would be set by the commands:

```
pop_name    1,height
type real,1
pop_xmin[1] =   0.0
pop_xmax[1] = 200.0
pop_smin[1] =   0.0
pop_smax[1] = 200.0
pop_sig [1] =   1.0
pop_lsig[1] =   0.1
adapt sigma , 1,0.2
adapt lsigma, 1,0.01
```

As of version 5.16, DIFFEV offers a shortened version of these commands with standardized values for the sigmas:

```
newparam height, 0.0, 200.0, 0.0, 200.0
```

If needed optional parameters can specify more details:

```
newparam height, range:[0.0, 200.0], value:[0.0, 200.0], status:free, init:keep
newparam height, range:[0.0, 200.0], value:[0.0, 200.0], status:free, init:initialize
newparam height, range:[0.0, 200.0], value:[0.0, 200.0], status:free, type:integer
newparam height, range:[0.0, 200.0], value:[0.0, 200.0], status:free, type:real
```

This command assumes the *good* values sigma = 0.001 and local sigma = 0.0001 and will adapt the sigma to a population width of 0.2 and the local sigma to 0.02 of the population width.
The command assumes that the parameter is real valued unless the optional parameter "type" is set to "integer".
Since parameter P_3 forms the denominator of Eq. 3.4, its value must not be equal to zero. Check the function with a couple of different parameters, or simple mathematical analysis would show that the product P_3 P_1 must also be larger than zero to produce the observed dip. A clear lower limit for P_3 can, however, not be given. Therefore, P_3 was allowed in the interval [-1:1], and we just have to exclude a value of zero.
If the problem is difficult to solve, or if one wants to get a quick estimate of one parameter, one can choose to refine just a subset. Since we only have three parameters, we will refine all at the same time.

```
constrain p[3].ne.0.0
```

The next group of definitions concerns the control variables. DIFFEV offers the two basic variables, the scale factor, by which the difference vector is multiplied, `diff_f[1]`, and the cross

over probability `diff_cr[1]`. Both are limited to the interval [0:1]. For this refinement problem, the actual value of the cross over probability does not matter, the scale factor should be closer to one to ensure successful refinement.
A variation of the basic algorithm allows to add the difference vector to any point along the line between parent and donor base. In this problem, the location does not influence the refinement, and we choose the value of 1, which corresponds to the original algorithm. Also, the value of the local search probability does not affect the search efficiency in this example, as long as its value is smaller than a value of roughly 0.3, and here it is set to zero.

```
diff_cr[1]  = 0.8
diff_f[1]   = 0.81
diff_k[1]   = 1.0
diff_lo[1]  = 0.0
```

The next choice concerns selection of the donor. One can either choose the current best member as donor, or choose the donor at random. Here the donor is chosen at random.
If the dependency of the R-value on the parameters is a reasonably smooth distribution without too many false minima, one can accelerate the convergence by taking the combined group of parents and children and to choose the best members to form the next set of parents. Here this is a good choice.

```
donor      random
selection  best,all
```

Finally we need to define filenames for the two log files. To keep the output nicely sorted, these files are written into a subdirectory `DIFFEV`.

```
logfile    DIFFEV/Parameter
summary    DIFFEV/Summary
```

This concludes the basic setup. Prior to the main refinement loop, we just have to generate the initial parameter sets, which is done by the command `initialize`, which does not take any parameters. This command sets the current generation number to zero and writes a first version of all trial files.

```
init
```

The main refinement follows, usually in a loop over several generations. Within the loop, the `system` command must be used to execute the slave program or programs that calculate the R-value. Once these are done, the `compare` command reads all R-values, and generates the next generation of trial values. It also updates the log files. For a fixed number of refinement cycles this could be a construction like:

```
do i[0]=1,40
  system ./arctan
  compare
enddo
```

A loop with 40 cycles is executed, using the external program "./arctan" to calculate the R-values. In a UNIX environment the leading "./" specifies that the program `arctan` is found in the current directory. Without this specifier, the UNIX shell would search for the program in the directories specified by the value of your *PATH* variable.
An indefinite loop that requires manual intervention could be:

```
 variable integer, terminate
 terminate = -1
 fopen 1, CONTINUATION
 fput  1, terminate
 fclose 1
 do while(terminate.eq.-1)
    system ./arctan
   compare
   fopen 1, CONTINUATION
   fget  1, terminate
   fclose 1
enddo
```

The macro initially sets the variable `terminate` to "-1" and writes this to the file `CONTINUATION`. The loop is executed until the value of the variable is no longer "-1". At each cycle the content of file `CONTINUATION` is read into variable `terminate`. Thus, the macro will stop, if you edit this file and change the number stored within.
Finally, the following macro checks the R-value and reacts accordingly. The commands in this macro were used to generate the performance tests in chapter 2.

```
variable integer,cycle
cycle = 0
do
  system ./arctan
  compare
  cycle = cycle + 1
enddo until (bestr[1].lt.0.0817830 .or. cycle.gt.100)
```

The loop is run indefinitely, until either the R-value has fallen below a threshold, or until the number of cycles exceeds 100.
A typical refinement run is shown in Fig. 3.4. The refinement is set according to the control variables described in this section. Initially, the best and worst R-values quickly drop to values around 9 to 12 %. In these cycles, those parameter values that are really far off are eliminated. Thereafter, the real refinement starts. Up to generation 37 little change occurs. At this stage, the first members have found the global minimum, and after generation 45 all members begin to converge into the global minimum, which is reached after generation 70. Thereafter, no significant progress occurs.
The main effect on the R-value is caused by the value of the parameter P_1, Fig. 3.5, since this parameter lowers or raises the function over the whole x range. Accordingly, the refinement quickly finds a value close to the right value. Notice that around generation 50, the parameter values spread before the final value is found. It is around these generations that all members find the global minimum and during the adjustment of parameter P_3, the other parameters spread out for a few generations.
For the first roughly 15 generations, the value of parameter P_2, Fg. 3.6, hardly affects the R-value. As long as the position of the minimum is not very close to the true value, its position hardly matters. From Generation 20 to 45 more and more members find the correct position.
Parameter P_3, Fig. 3.7, shows the most unusual refinement behavior. In the first few generation, all members with negative values of P_3 are eliminated. The parameter then refines to values between zero and 0.02, much lower than the true value. As long as the position of the minimum is not found, lower R-values result if parameter P_3 is as close to zero as possible. To get out of this local minimum, a large population size is required. Once the correct position is found, parameter P_3 also refines to its correct value.

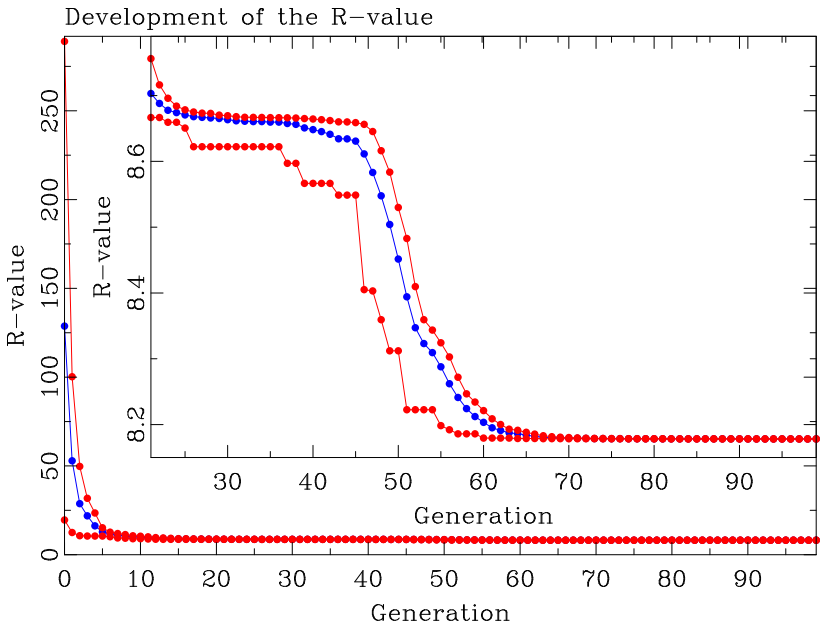


Figure 3.4: R-value as function of refinement generation. The Figure shows the best, worst R-value (red curves) and the average R-value (blue)

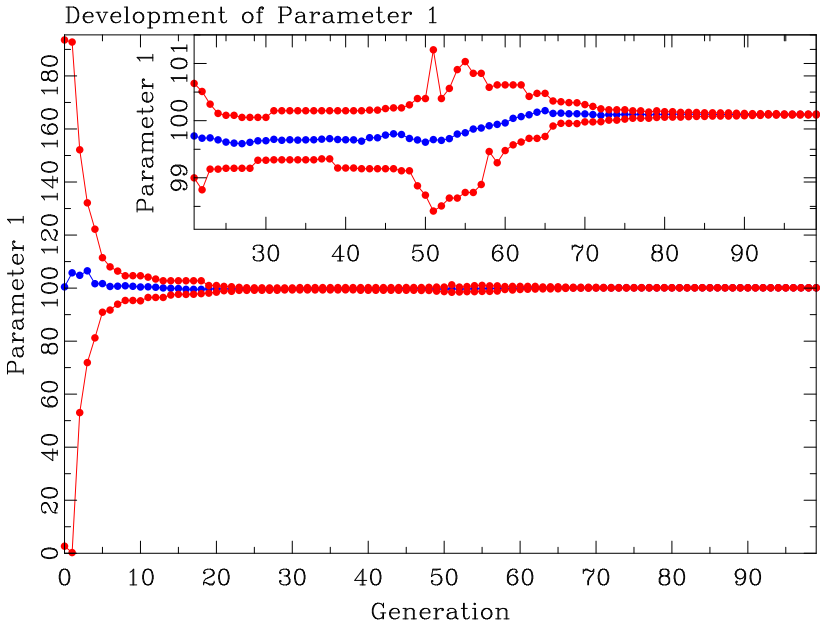


Figure 3.5: Parameter P1 as function of refinement generation. The Figure shows the best, worst R-value (red curves) and the average R-value (blue)

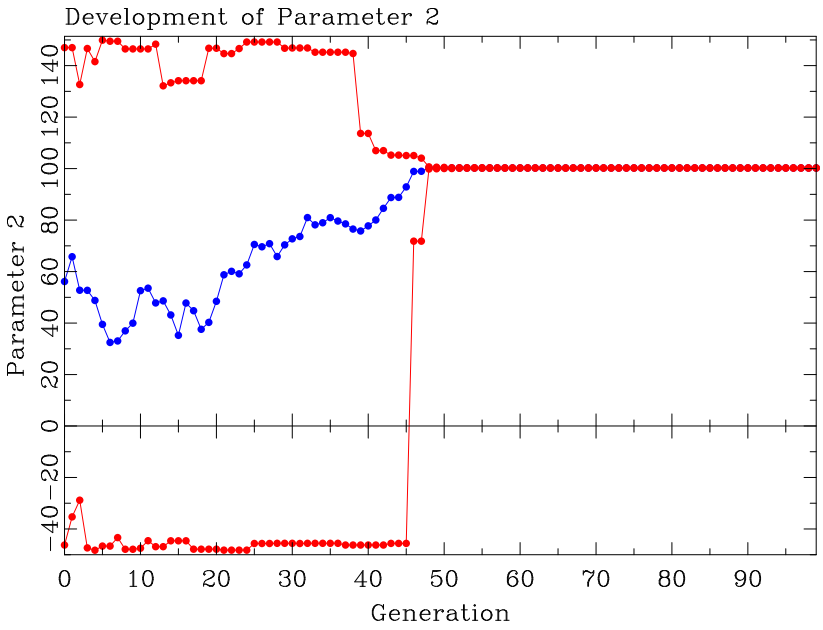


Figure 3.6: Parameter P2 as function of refinement generation. The Figure shows the best, worst R-value (red curves) and the average R-value (blue)

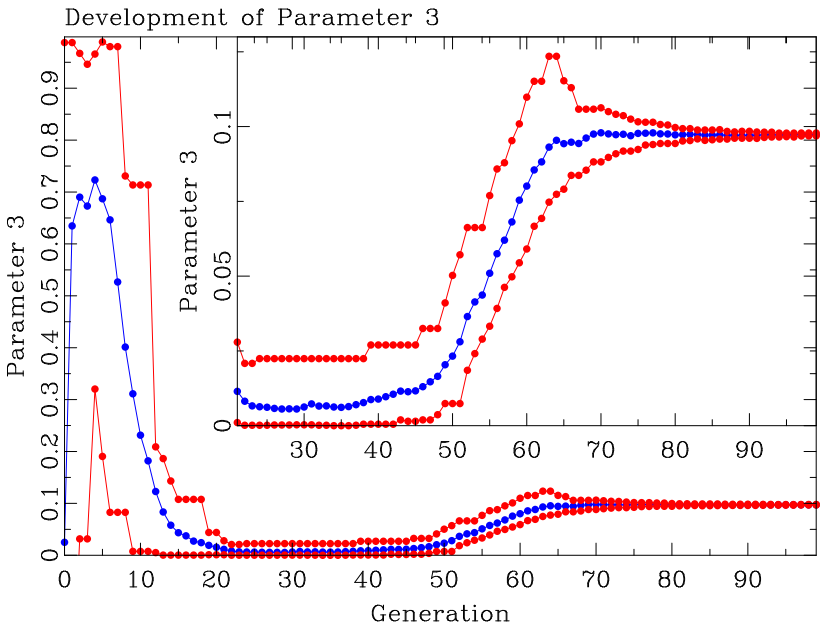


Figure 3.7: Parameter P3 as function of refinement generation. The Figure shows the best, worst R-value (red curves) and the average R-value (blue)

At this stage, the refinement is finished and the calculated curve is that of Fig. 3.2. Admittedly, the difference between the R-values around generations 30 at 8.65% are not very significantly worse that the final R-value of 8.18%. This is due to the large amount of noise in the *experimental* data. These conditions were, however, chosen on purpose to illustrate the ability of the differential evolutionary algorithm to jump out of local minima, even under adverse conditions.

3.3 Fixing, Releasing refinement parameters

Most of the time, one will want to refine the parameters that have been defined and given a range with the `newpara` command. This was done in the macro `diffev_setup.mac` in section 3.
There are, of course, situations when you might want to fix one or several parameters. This can be done with the DIFFEV command `fix` that takes the form:

```
fix <Parameter_name>, <value>
fix <Parameter_name>, best
```

The first command form fixes the parameter called `Parameter_name` to the number given by `value`. The second form fixes the parameter to the value that correspond to the current best member in the population.
Instead of the parameter name, the parameter number can also be given, but the preferred syntax would be to use the parameter name.
The `fix` command sets the refine glag for the parameter off. The command also sets the absolute and the starting boundaries to the parameter value.
The complementary command to `release` a fixed parameter takes the form:

```
release <Parameter_name>, range:<sigma>
```

This command initializes the parameter to a range `current value ± sigma`. The lower and upper absolute parameter boundaries are fixed to `current value ± 3*sigma`. The refine flag is set to `refine`.
To fine tune the `release` command the following further optional parameter are available:

```
value:<setpoint>
min:<pop_xmin>
max:<pop_xmax>
dismiss:<number>
```

The `value` parameter serves to define a new setpoint for the parameter insteiad of the current (fixed) value. With `min` and `max` you have full control to define the lower and upper absolute parameter boundaries.
Take as an exampe, the stacking fault parameter from the example in section 3. It could be a good idea to constrain this parameter to a fixed value of 1.00 at the beginning of the refinement. This would correspond to an ideal zincblende structure free of stacking faults. Later on the parameter might need to be refined. As the DISCUS stacking fault parameter is fixed to a range 0 to 1, the release command is well off to set a new setpoint at a value somewehere between 0 and 1. If we take as an example a setpoint of 0.9, and a range of 0.4, DIFFEV would initialize the parameter in a range from 0.5 to 1.3. Furthermore, the absolute limits would be set to -1.1 and 2.1. Most of these numbers are outside the physical limits of the DISCUS stacking fault

parameter. Thus it is necessary to set the limits explicitly to 0.0 and 1.0 or other values within the allowed range.
If a parameter is released, chances are, that the R-values of all children are worse than the parents. In this case all trial parameters for the children would be ignored and DIFFEV will take the parent values to generate the next generation. In this case, all parent values for the parameter that we want to release would revert back to the constant fixed value and the release would be lost. To prevent this from happening, DIFFEV offers the main command `dismiss` or the optional `dismiss:` parameter at the `release` command. The purpose is to assign to some of the parents an excessivly high R-value. This will be done to the worst members in the population and ensures that for each of these members the child will have a lower R-value and will thus survive into the next generation. The `dismiss:` parameter expects the number that signifies how many of the worst members shall be dismissed. The number can be specified either as a constant value or an expression. The allowed range is between 0 and the number of members in the population. With a value of "1", only the worst member is dismissed, with "2" the two worst members and so on. Otherwise the number can be given as keywords "all", "best" or "none".

keyword	Description
0	No member of the population is dismissed
n	The worst "n" population members are dismissed
all	All members are dismissed
best	All but the best member are dismissed
none	No member is dismissed, same as number 0

Chapter 4

Creating a DIFFEV Macro Set

Writing all the required macros from scratch can easily be an error prone task. To ease this step the DISCUS SUITE includes an easy setup for a refinement template. The template can be build using a SHELX instruction file as the initial input. Such a file will provide a full set of information regarding a single crystal refinement, yet it can also be used to define at least a generic template for a powder diffraction and a Powder Pair Distribution PDF refinement in DIFFEV. As a SHELX instruction file does not contain any information regarding a diordered structure you will still have to do some editing of the macros that will have been generated with the setup procedure. You will, however, be able to start from an working example.
The DISCUS SUITE is not primarily designed to perform a classic single crystal or Rietveld type refiment. Specialized single crystal refinement programs like SHELXL respectively Fullprof, Topaz, Maud, GSAS-II etc. are certainly much better suited to perform this task. This setup is intended as a template to expand the initial refinement into a full refinement of a disordered structure.
In this introduction, the common set up scheme is explained. The individual sections that follow will focus on the individual details and in particular will explain which macros may have to be adapted for the refinement of a disordered structure.
All of the following sections assume that a SHELXL instruction file exists. If you do not have such a file at hand, DISCUS can generate the file with the instructions in this short macro.

```
1  discus
2  read
3  stru CELL/znse.cell
4  export shelxl, znse_wurtzite.ins, cycle:10, lambda:MOA1
```

The macro reads the asymmetric unit of the Wurtzite type modification of ZnSe and exports this information into a SHELXL instruction file. The optional parameter `cycle:10` instructs the instruction file to perform 10 refinement cycles. A further option is to provide the wave length as symbol or numeric value. Make sure that you read the asymmetric unit only.
All of the setups in the next four sections use a macro line within the DISCUS section equivalent to:

```
1  import shelx, znse_wurtzite.ins, names:shelx, diffev:[style:single, form:waas, &
2         compute:serial, lambda:MOA1]
```

This command expects to find a file `znse_wurtzite.ins` in the current directory. The `import` command will read the SHELX instruction file and convert it into a discus `cell` file, here using the actual SHELX atom names.
The optional parameter `diffev` and its parameters instructs DISCUS to set up a DIFFEV refinement scheme. See the REFINE manual for an equivalent set up for the REFINE section.
The `style` parameter defines what type of refinement should be created

style:single	Run a single crystal refinement. DISCUS will look for an `hkl` or `fcf` file in the current directory. If either exists a Bragg reflection setup will be created, otherwise a 3D diffuse scattering setup.
style:powder	Run a powder diffraction refinement.
style:pdf	Run a PDF refinement.

The `form` parameter lets you define the atomic form factors to be used:

form:table	Use atomic form factors as tabulated in the International Tables Vol C
form:waas	Use atomic form factors as tabulated in Waasmeier and Kirfe. This is the default.

The parameter `lambda` lets you define the wave length for the refinement. If absent, the wave length defaults to the value found in the SHELXL `CELL` instruction.
The `compute` parameter defines if inidividual repetions are to be calculated in parallel or serially:

compute:serial	Compute the individual repetions serially within the same DISCUS macro, Default
compute:parallel	Compute the individual repetitions in parallel on separate compute nodes.

The setup generates a macros `setup_rep.mac` that contains a single line:

```
1 REF_NINDIV = 1
```

The number of individual repetitions default to 1 (serial) and 2 (parallel). Adjust these as needed.
All styles will create a set of required directories:

CELL	Will contain a copy of the cell file, the refinments read this copy. The macro at the beginning of this chapter read a cell file from the CELL folder. Such a cell file will be replaced by the result of the `import` command!
DATA	Place the observed data into this directory. An old directory is not changed.
DIFFEV	The main directory for all DIFFEV archive files
FINAL	The place for the current best calculated data
INDI	The place for (temporary) individual calculated pattern
LOGFILES	Place for temporary log files that are created if DIFFEV is run in parallel
TEMP	Averaged individual calculations are placed into this directory

All styles generate a macro `diffev_main_SUBSTANCE.mac` that needs to be used as the main refinement macro. Here and in all following macro lines the string tt SUBSTANCE should be understood as being replaced by the name of the SHELXL instruction file, in these examples it would be `znse_wurtzite`. At a Linux of MacOS operating system you can perform the refinement with the command:

```
mpiexec discus_suite -macro diffev_main_SUBSTANCE.mac
```

At Windows run the discus_suite from the icon and perform a parallel refinement with

```
parallel diffev_main_SUBSTANCE.mac
```

To test the initial macro, especially when you need to add code to perform a simulation of a disordered structure the setup provides an initial version of `diffev_best.mac`, which can be run at the suite prompt. Keep in mind that this macro will be replaced by updated versions during the refinement.
Equivalent to the main DIFFEV macro you will find the continuation `diffev_continue_SUBSTANCE.mac` to perform further cycles. Both the main and the continuation macro will perform only 10 cycles and use a very restricted set of refined parameters and a tiny population size. You will need to adjust these to your individual need, see `diffev_setup.mac`. In `diffev_setup.mac` DISCUS defines the refined parameter names as `P_the_name` and `U_Atom_N` for the anisotropic atomic displacement parameters. These parameter names are not fixed, feel free to adjust to your own style. During the setup DISCUS checks the SHELXL instruction file and defines as parameters only any atom coordinate that is not fixed by symmetry. Initially all atom coordinates are fixed and you need to free them as needed.
As an example the following macro is generated for single crystal Bragg intensity refinemnts:

```
 1 variable integer, diff_counter
 2 diffev
 3   reset
 4   data hklf4, DATA/znse_wurtzite_merged.hkl ! Load data file, populates F_XMIN etc.
 5   @cleanup.mac       ! Remove old files
 6   @setup_rep.mac     ! Define number of individual repetitions
 7   @global.mac        ! Define global file and directory names
 8   @diffev_setup.mac  ! Define refinement details
 9   init               ! Initialize all parameters
10   do diff_counter = 1, 10  ! Perform 10 loops!
11     echo "In loop %4d ", diff_counter     ! Keep user posted
12     run_mpi discus, dis.diffev.mac, repeat:REF_NINDIV, compute:serial, logfile:LOGFILES/d
13     compare
14   enddo
15 exit   ! Back to SUITE
16 exit   ! Terminate SUITE
```

This macro is very similar to the initial macro in chapter 3. Two aspects need to be mentioned here.
The `data` command in line 4 will load the data file from folder DATA. DISCUS does an educated guess to the file format and name to be expected in this directory, adjust as needed. At the moment DISCUS reads the file with the sole intention to populate variables `F_XMIN, F_XMAX, F_XSTP` and equivalently Y and Z for 3D data.These limits are subsequently used in the macros that calculate / write the calculated data.
Command `@global.mac` (line 7) runs a macro that defines variable names and values for the data directory, the location of individual repetitions, temporary structure data and last but not least the substance itself.

4.1 Setting up a single crystal Bragg refinement

With `style:single` DISCUS will look for files `SUBSTANCE.hkl` and `SUBSTANCE.fcf`. If either is found, a single crystal refinement using the integrated Bragg intensities in these files will be created. As the `fcf` file contains a merged set of the Bragg reflection intensities the refinement will be faster. Without modifications the refinement will replicate a SHELXL single crystal refinement. If the `fcf` file is present, DISCUS converts this into an hklf4 format under the name `SUBSTANCE_merged.hkl`
In the initial setup all parameters except the scale factor `P_scale` are fixed within the parameter block in macro `diffev_setup.mac`. To free any parameter, edit this file and add a line `refine P_Name` for the corresponding parameter name to macro `diffev_fix_free.mac`.
The `data` command in `diffev_main_SUBSTANCE.mac` will load the hkl file with the hklf4 type.

4.2 Setting up a single crystal refinement

If the current directory does not contain an `hkl` or `fcf` file, DISCUS will assume that a 3D diffuse scattering calculation is intended. You need to make sure that the DATA directory contains a suitable file. If the observed data are not in the hdf5 file format, several macro files: `diffev_main_SUBSTANCE.mac; diffev_continue_SUBSTANCE.mac; diffev_best.mac; global.mac; ksingle.mac; kup.diffev.mac; SUBSTANCE_main_single.mac` need to be modified.
In the main DISCUS macro `SUBSTANCE_main_single.mac` check the commented lines

```
1 !set filter, lanczos, damp:0.5, width:7, scale:1.0
2 !set symmetry, apply
3 run ! sigabs:[0.001, 1.0,0.0,0.0, sigord:[0.001, 0.0,1.0,0.0], sigtop:[0.001, 0.0,0.0,1.0]
```

Do you need to smooth the data with a filter (line 1), is symmetry averaging allowed (line 2) and does it help your calculated data. Lastly do you need to convolve the experimental data with a resolution function (comments on the run command, line 3).
Check if the `nufft` technique works fine for your sample.
Another likely modification is to switch the calculation to a $3D$-Δ-PDF. Besides the observed data you will need to modify the output in the main DISCUS macro `SUBSTANCE_main_single.mac` to something like:

```
 1 fourier
 2   ...
 3   set aver, 100
 4   ...
 5   run
 6 exit
 7 output
 8   outf "%c/INDI/indi.%4D", INDIDIR, REF_KID
 9   form hdf5
10   value 3DPDF
11   run
12 exit
```

With line 3 `set aver, 100` a total of 100 percent of the actual crystal is sampled to calculate the Bragg intensities, which are subtracted from the calculated diffraction pattern. This creates the $3D$-Δ-PDF instead of a 3D-PDF.
A $3D$-Δ-PDF makes sense for an extended simulated crystal with disorder only. You will have to expand the crystal to a larger block of unit cells in the main DISCUS macro where the initial internally saved asymmetric unit is expanded at line `cell internal.SUBSTANCE.cell` and introduce the required disorder.

4.3 Setting up a powder diffraction refinement

With `style:powder` DISCUS will create a DIFFEV refinement for powder diffraction. The variable set and the powder macro will contain fixed parameters for the profile function and comments to assist in the setup for preferred orientation. Initially the scale parameter and the lattice parameters are free, all atom positions, thermal parameters and a 2Θ zero point are fixed. The `data` command and the output in the main substance macro assume a 2Θ scale.
In macro `kup.diffev.mac` multiple individual repetitions (if requested) are merged and a background polynomial is fitted. The sum of the background and the scaled calculated powder pattern is saved and serves as the total calculated data set.

4.4 Setting up a PDF refinement

With `style:pdf` DISCUS will create a DIFFEV refinement for the powder pair distribution PDF. The generated macros are mostly identical to the powder case. The main difference is that the automatic setup assumes an ellipsoidally shaped nanoparticle. Accordingly the powder pattern is calculated via the Debye-scattering-equation (DSE) assuming a wave length of 0.2Å and $Q_{max} = 26$Å. As the SHELXL instruction file does not contain any information on a nanoparticle shape or size, dummy values for the three diameters are used. These need to be adjusted for your own data.
The `data` command assumes that the observed PDF is present as a four column file with columns `r PDF 0 sigma(PDF)` with equidistant r scale. Adjust if needed. The limits and step sizes are copied into the calculated PDF.

4.5 Adding disorder to a refinement

Disorder within a simulated crystal is very much an individual problem. Consult the examples and algorithems in the DISCUS manual and the Cook book for general guidelines. The initial Nanoparticle example in the chapter 3 details the simulation of a finite sized nanoparticle with stacking faults.
For a single crystal refinement (diffuse or $3D$-Δ-PDF) keep the simulated crystal to a block shaped object, see the notes on the crystal size in section 4.2. This facilitates periodic boundary conditions and enables a fast internal lookup that is often needed for example within the Monte-Carlo-menu `mmc`.
An important design choice is the question whether the disorder is represented within a single simulated crystal or if many individual crystals are required to obtain an ensemble average that

corresponds to the experimental situation. Often, a suitably large single crystal will be a good represenation, especially if the correlations between the structural defects do not extend over many unit cells. A good example of the latter is the first example in chapter 3. A single very small ZnSe cannot represent the multitude of nanoparticles with different number and location of the stacking faults. Many particles must be simulated and their averaged powder diffraction pattern or PDF will be a good represenation of the actual experimental situation.
If no individual repetitions are needed, or if all of them should be calculated on the same CPU use the `compute:serial`. This is a good choice if the computation time for each simulation is pretty much the same. All members of the population will take the same time and not too much time is lost waiting for a single slow member to be finished before DIFFEV can move on to the next `compare` command and the next generation.
If the computetation time for members or for individual calculations of the same meber do vary considerable it is etter to use the `compute:parallel` parameter. In this case individual repetitions for all members are distributed over all CPUs and a more even workload for each CPU can be expected. This minimizes the delay time at the end of a generation. See section 2.5 for full details. Try out the setup generated by both variants in order to determine the best solution. parameter.

Appendix A

DIFFEV commands

A.1 News

2019_Jun

If a slave process terminates with an error message, a new macro is written that is essentially identical to "diffev_best.mac". This macro called "diffev_error.NNNN.NNNN.mac", where "NNNN.NNNN" are the REF_KID and REF_INDIV number of the slave process that failed. If this macro is run in a regular DISCUS_SUITE session it will recreate the error with exactly the same parameters and random numbers as the failed slave.
A bug was fixed in the handling of the random state for "diffev_best.mac". The random state is now recorded exactly.

2019_Jan

Improved the writing of the diffev_best.mac macro. The state of "compute:" is evaluated and multiple "run_mpi" lines are taken into account.

2018_Nov

Added a new command 'restart <user_generation>' This command compares the user value to the current value in GENERATION. If the user value is smaller, the Logfiles, Summary and Last files are shortened and the GENERATION file is adapted. This lets you step back to an earlier refinement status. Keep in mind that the later cycles are irrevocably lost. If in doubt make a backup first.
reinstated the ==> 'read' command
Added further options to the ==> 'release' command

2018_Oct

Added a ==> 'release' command that is intended to act complementary to a previous ==> 'fix' command.

2018_July

A small modification to the "diffev_best" macro. The value of REF_NINDIV is written and at the end a "set error, continue' instruction has been added.

2018_June

Revised the reaction to a CTRL-C
Added a ==> 'set error, ... , "save" option
Revised the internal workings of the distribution within run_mpi. This has no effects on the user. It should enable you, however, to distribute the children much better on a system with many nodes and many CPU's per node. See the manual for further info.

2018_Feb

Added an optional parameter "partial:<value>" to the ==> 'restrial' command. This parameter tells DIFFEV how many partial R-values the slave program will calculate and return. If set or if the value is larger than 1, DIFFEV will create logfiles "Summary.Rvalue.0001", "Parameter.Rvalue.0001" and "Current.Rvalue.0001" etc. for each partial R-value. Use the KUPLOT commands 'rval' and 'cost' to set the individual R-values and a (weighted) average. You can display the development of the partial R-values within KUPLOT" with the kuplot command 'kpara'.
removed 'read' command

2018_Jan

The logical comparisons may now take the operators: <, <=, ==, /=, >=, >/ The classical fortran77 operators are still valid
New logical functions "isvar" and "isexp" can be used within an "if" construction. See help entry ==>'function' in the general "Command_lang" section
The ==> function par_number(<char_variable>) returns the number for the refinement parameter in the character variable <char_variable>.
The ==> function par_name(<number>) returns the name of the refinement parameter number <i>.
Finished the transformation from parameter numbers to parameter names. All Log files now have an extension of the parameter name instead the parameter number.
The new command ==>'reset' can be used to reset DIFFEV to the conditions at program start.

2018_Dec

Major revision of the refinement parameter handling. The new command 'newparam' effectively replaces for most of the common refinements the need of the individual commands to set the parameter values. Throughout the command language much more emphasis is placed on the parameter names instead of the parameter numbers. See ==>'refine' and 'init' commands. The parameter names are also placed into the user variable environment and handed

down to the slave program (discus or kuplot) where they will have the appropriate values for the members of the population.
Added a 'read' command that reads GENERATION and the ==> 'logfile' This allows for easier changes of the population size, see the entry 'Description / Population_size'

2017_Sep

Throughout the program the internal calculation of random numbers was changed to the FORTRAN 90 intrinsic function.

2017_July

DIFFEV has been modified to log the status of the random number generator at the beginning of each slave calculation. This status is documented internally for the current best member of the population. Once an 'exit' command is executed, DIFFEV will write a macro called "diffev_best.mac" that can be used to recreate the current best solution.

2017_Jan

An unfortunate typing error in News/2016_Oct regarding the new refinement variable ref_para[1...] (was misspelled as ref_param[1...]) is corrected in the on-line help.
Another typing error caused an error in the macro parameters transferred with a NON MPI command: run_mpi, making these not backward compatible. This has been fixed.

2016_Dec

At a few select points colors are introduced into the output. Currently these are just the error messages.

2016_Oct

Global variables have been introduced that use the same syntax as user defined variables. This include just "pi" and variables related to the refinement. DIFFEV sets the value to these variables: REF_GENERATION Current generation REF_MEMBER Current population size REF_CHILDREN Current children size REF_DIMENSION Number of parameters REF_KID Current child Updated for DISCUS and KUPLOT only REF_INDIV Current individuum Updated for DISCUS and KUPLOT only REF_NINDIV Number of individual repetitions ref_para[1..] Current trial parameters for current child

2016_june

DIFFEV may now be interrupted gracefully with a CTRL-c. This will cause DIFFEV to shut down MPI if active.
The 'run_mpi' command can now be used as a generic interface to identical slave macros, regardless of the MPI status.

2016_april

The 'initialise' command was augmented by a second form to initialize just the logfiles, see ==> 'logfile', 'summary'. This might be helpful if you want to reset the refinement cycle to a smaller generation number. For very lengthy refinements with a few thousand refinement cycles and many parameters, a continuation will take appreciable time to read all previous cycles. If the actual development across the cycles is not relevant to you you can reduces the file sizes drastically be a sequence like: pop_gen[1] = 1 initialise logfile
New command 'lastfile'
This new command creates a short copy of the ==> 'logfiles'. This short form contains the parameters just for the last refinement generation.

2013_May

Sockets have been removed, this comment is obsolete.
The run_mpi command has been augmented by a "socket" qualifier, which will cause DIFFEV to run the application program controlled via a socket. This should speed up the process on a multi-core/multi-cpu system.

2011_June

The program is now a fully functional fortran2003 program. There are no changes that must be followed by the user, but important differences exist in the memory allocation. The internal arrays that hold the population are now all allocated automatically, when you define the number of parameters to be refined, the size of the population and the constraint equations. Most of the time you will not have to bother with these computational details. If you wish, you can allocate appropriate array sizes, see ==> 'allocate', 'deallocate'.
With this version, the program allows the user to change the population size and the number of parameters to be refined during a refinement cycle.
If the number of parameters to be refined in increased during a cycle, the program will automatically write new child parameter sets and patch the logfile and summary file.
See the entry ==> "Description" ==> "Increase_Dimension" for an example.
Related to this new feature, the ==> "initialise" command now allows to (re-)initialize an individual parameter.

A.2 Synopsis

```
Description ! A description of the program
News        ! Information on recent changes
allocate    ! Allocate array sizes
adapt       ! Adaption of global/local search width
backup      ! Backup current best solutions
compare     ! Compares the results of the current population
constraint  ! Defines a constraint condition
deallocate  ! Deallocate array sizes
dismiss     ! Dismiss the worst parents; replace by new children
donor       ! Defines which donor to use
fix         ! Fixes a parameter
initialize  ! Initializes the generation zero
```

```
lastfile    ! Defines a file name for the parameters for the last generation
logfile     ! Defines the file name for the parameters for each generation
newparam    ! Defines Ra new parameter or new values for a parameter
purge       ! Purges old Generations from log files
pop_name    ! Defines names for the individual parameters
refine      ! Defines which parameters are refined/fixed
release     ! Initialize a parameter that had been fixed
restrial    ! Defines the file name for the current R-values
run_mpi     ! Run the cost function program in parallel through MPI
selection   ! Defines how children/parents survive into next generation
summary     ! Defines the file name for a summary of the parameter changes
trialfile   ! Defines the file name for the current parameters
type        ! Sets the numerical type for a parameter (integer or real)
write       ! Writes new Children or new GENERATION file
```

A.3 Description

The program DIFFEV uses the differential evolution algorithm to refine a set of parameters to a set of observations.
DIFFEV provides the handling of the parameters and their evolution. An external program must be used to calculate the cost function or R-value that corresponds to a given set of parameters.
The differential evolution algorithm compares simultaneously the resulting cost function for several sets of parameters. The number of parameter sets is called the population. For each member of the population a set of parameters is used. This is called a parameter vector. Its dimension depends on the model that you need to describe. In order to describe a parabola you would need three parameters. The whole set of parameters is called a generation.
For each generation, the cost function is evaluated for each member of the population. The next generation is determined from the current parent generation through the following procedure: Loop over all members - Choose a member (at random or at will), this will be the donor base. - Set a point along the line between current member and donor base to create the effective donor base ==> diff_k[1] - Select two other members by random choice - Loop over all parameters: - Take the difference between the corresponding parameters of the two other randomly chosen members. - Multiply this difference by a user provided value ==> diff_f[1]. - Add the difference vector to the effective donor base to create a parameter set called the donor. - One parameter of the donor is always chosen for the child. All other parameters are then randomly chosen from: Either: the parent Or : the donor - The probability for this choice is weighted by a user provided probability ==> diff_cr[1].
Once a new generation has been determined the corresponding cost function is calculated.
Next the selection process determines, which current children survive into the next generation. This is done either by: - Direct comparison between a parent member and its immediate child. Only those new members survive, whose cost function is less than that of the parent. Otherwise the parent is retained. - All parents and all children are pooled into one set. From this set the best members survive, irrespective whether they were a parent or a child.
Such an algorithm is able to search for parameters if a standard refinement algorithm fails or is difficult to adapt. This might be the case for: - Undefined parameter values within the possible range. The parameter P may for example NOT be equal to 1 for a function: 1/(1-P) but values larger AND smaller are allowed. - The calculation of the cost function involves existing

extensive algorithms or several different programs. - The calculation involves the averaging of several calculations that rely on data created by (Gaussian-) randomly distributed parameters.
Files
DIFFEV uses several files to store the parameter values, the current status etc these are:
"GENERATION" Fixed file name in the current directory!
The file "GENERATION" contains twelve fixed lines with the current generation number, and all relevant file names. Further lines contain the backup file names if defined, the random number seeds and the finally the parameter names and their allowed numerical ranges.

```
# generation members children parameters
     158        45        90         4
# trial file
DIFFEV/Trials
# result file
DIFFEV/Results
# log file
DIFFEV/Parameter
# summary file
DIFFEV/Summary
# last    file
DIFFEV/Current
```

Trial files (obsolete)
DIFFEV writes a short file for each member that contains the current parameters for one member. The user defined file name is appended by a four digit member number. The first lines contain the information on current generation number, the number of members in the population, the number of children and the number of parameters

```
# generation members children parameters
     181        45        90         4
# current member
    1
# parameter list
    0.8284027688E-02
    0.5815573883E+02
    0.2050311089E+01
    0.3000000000E+01
```

Result files (obsolete)
DIFFEV expects to read the R-values for each member from a short file. The user defined file name is appended by a four digit member number.
The file must contain the member number and the R-value in free format within the first line.
Parameter files
For the R-value and for each parameter a SPEC type file is written that contains all old R-values and parameters, respectively. Each generation makes up a scan. The first column is the member number, the second column is the R-value and the third column is the respective parameter value. The base name of the parameter files can be set by the user via command ==> 'logfile'. As of Version 5.16.1 the base name is appended by an extension that is identical to the parameter name. Parameter file <name>.Rvalue has the same structure, except that both column two and three are the R-values.
Prior to version 5.16.1 the base name was appended by a four digit number with leading zeros. Parameter file no 0000 has the same structure, except that both column two and three are the R-values.

Last file
A short copy of the parameter file that contains the parameters just for the last generation.
Summary files
For the R-value and for each parameter a SPEC type summary files contains a single scan. Each generation creates one line within the scan. Five values are written to each line. The first column is the generation number For the R-value and each of the parameters four further columns are written. The first of these is the average value, the second the minimum value, the third the maximum value and the fourth the sigma of the parameter distribution. The base name of the summary files can be set by the user via command ==> 'summary'. This base name is appended by a four digit number with leading zeros. Summary file no 0000 has the same structure, except that both column two and three are the R-values.
Further help topics are:

Basic_Example

This example illustrates the commands to refine the three parameters that describe a parabola: y = P1*x**2 + P2*x + P3
You will find the data and the macros in the diffev/Example directory within the program source directories.

```
======================diffev.mac=======================================
#
#
pop_gen[1]  = 0         # initialize the current generation to zero
#
pop_n[1]    = 15        # The population shall have 15 members
pop_c[1]    = 15        # The population shall have 15 children
pop_dimx[1] = 3         # We need three parameters
#
pop_name      1,square  # The parameter is called "square"
#
type          1,real    # Parameter 1 is a floating number
#
pop_xmin[1] = -1.0      # The first parameter is restricted to the
pop_xmax[1] =  1.0      # range -1 to +1
#
pop_smin[1] = -0.8      # The starting parameters of the first parameter
pop_smax[1] = +0.8      # are restricted to the range -0.8 to +0.8
#
pop_sig[1]  =  0.02     # The sigma of minimum parameter "noise".
pop_lsig[1] =  0.002    # The sigma for local searches
#
adapt  sigma, 1, 0.025  # After generation 0 the value of pop_sig[1]
#                       # is adjusted to 0.025*(largest parameter -
#                       #                       smallest parameter )
#
adapt lsigma, 1, 0.0025 # After generation 0 the value of pop_lsig[1]
#                       # is adjusted to 0.0025*(largest parameter -
#                       #                        smallest parameter )
#
pop_name      2,linear
pop_xmin[2] = -2.0
pop_xmax[2] =  2.0
pop_smin[2] = -1.8
pop_smax[2] = +1.8
#
pop_name      3,constant
```

```
pop_xmin[3] = -5.0
pop_xmax[3] =  5.0
pop_smin[3] = -4.8
pop_smax[3] = +4.8
#
constraint  p[1].lt.1.3 # Several constraint may be imposed on the
constraint  p[2]+p[3].gt.0.0
#                       # parameters. Here the first parameter must be
#                       # less than 1.3. In the second constraint condition,
#                       # The sum of the second and third parameter must be
#                       # greater than zero.
#
diff_cr[1]  = 0.9       # The cross over probability is 90%
diff_f[1]   = 0.81      # The difference vectors are multiplied by 0.81
diff_k[1]   = 1.0       # For diff_k = 1, the difference vector is added
#                       # to the donor, for diff_k = 0 to the parent.
diff_lo[1]  = 0.1       # In 10% of all cases, a member creates its child
#                       # not by diffev algorithm, but from a local Gaussian
#                       # distributed search.
#
trialfile  silent       # Instead of writing to a file DIFFEV will pass the
#                       # values silently down to DISCUS/KUPLOT
restrial   silent       # Instead of reading a resultfile, DIFFEV will
#                       # obtain the cost function results directly from
#                       # the slave program.
logfile    Parameter    # The log file for the parameters
summary    Summary      # A shorter log file
#
init       silent       # Initializes diffev, Generation zero is written.
#
do i[1]=1,15            # A loop over 15 generations
  run_mpi kuplot, kcompare.mac, 0,  /dev/null
#                       # diffev starts the kuplot program with input
#                       # from file "kcompare.mac". This macro instructs
#                       # kuplot to read the 'trialfiles', to calculate the
#                       # cost function for each member and to write the
#                       # results into the 'restrial' files.
#
  compare  silent       # diffev reads the 'restrial' files, compares the
#                       # cost functions of children and parents and
#                       # creates the next generation.
#
enddo                   # End of the loop
```

Increase_Dimension

Here is an example for a macro that should be used to increase the number of parameters to be refined:

```
newparam cube, -5.0, 5.0, -4.8, 4.8, keep:initialize
#                               # Defines a new parameter called "cube"
#                               # that is allowed to be in
#                               # the range [-5:+5]
#                               # It is initialized in the
#                               # range [-4.8:+4.8]
dismiss          pop_n[1]/2     # Set R-value of half the population
                                # to a very high value, thus they will
                                # be replaced in the next generation
```

Prior to version 5.19 the macro would have taken the following command:

```
variable integer, ipar           # just a nice variable name
pop_dimx[1]    = pop_dimx[1] + 1 # increase dimension
ipar           = pop_dimx[1]     # copy into variable

pop_name         ipar,cube       # Define name etc for new parameter
pop_xmin[ipar] = -5.0
pop_xmax[ipar] =  5.0
pop_smin[ipar] = -4.8
pop_smax[ipar] = +4.8
type real,       ipar
refine           ipar            # Set refinement flag
init             ipar            # Initialize just this new parameter
dismiss          pop_n[1]/2      # Set R-value of half the population
                                 # to a very high value, thus they will
                                 # be replaced in the next generation
```

Hints

These are some hints regarding useful parameters. They are derived from the authors experience and are to be carefully adopted.
The population size MUST be at least 4!
The population size should be at least about ten times as large as the number of parameters, twenty times will give you a good sampling of the parameter space. A smaller population runs the risk of converging into a local minimum.
The cross over probability ==>diff_cr[1] should be about 0.8 A smaller value (lets say around 0.3), seems to prevent convergence, While a small value prevents the special properties of the differential algorithm to be applied at all. At diff_cr[1]=0, all children would always be identical to their parents!
The multiplier for the difference vector ==> diff_f[1] should be around 0.8 A small value prevents the children from being very different from their parents, while a large value > 1.5(?) seems to prevent convergence, since all children always jump too far away from their parents.
selection mode If the dependency of the cost function/R-value on the parameters is (or seems to be) fairly straightforward, the convergence is much faster if you choose the selection best,all scheme. By increasing the number of children in comparison to the parents, the selection pressure also increases and the algorithm will move faster into the minimum. This will, however, also happen if you happen to be close to a local minimum, instead of the global minimum!
I cannot give a hint regarding the number of generations required. This depends too much on the problem at hand, parameter correlation, the initial choice of parameters etc.

Population_size

Usually the population size will remain fixed during the course of a refinement. The population size is defined by setting appropriate values to the variable 'pop_n' and 'pop_c' for the parent and children respectively.
If you want to change the population size during a refinement, use a sequence of commands as follows:
read ! This forces DIFFEV to update the parameters from ! GENERATION and the logfiles pop_n[1] = <new_value> pop_c[1] = <new_value>
You may increase or decrease the population size as needed.

A.4 manual

```
manual ["section:"{"suite"  | "discus"  | "diffev" |
                   "kuplot" | "package" | "mixscat"}
       [,"viewer:"<name>]
```

Opens a PDF viewer for one of the Manuals
The section defaults to the current program section that you are working with. On Linux systems, the viewer defaults to "qpdfview", on Windows system it defaults to "firefox". If DISCUS does not find the default or the user provided viewer, DISCUS will search a list of common PDF viewers. If none is found an error message points to the folder that contains the manuals.

A.5 allocate

```
allocate
allocate "default"
allocate "constraint", <max_constr>
allocate "population", <max_members>, <max_parameters>
allocate "show"
```

DIFFEV allows to to allocate memory for the arrays needed to store the population and the constraints. DIFFEV will dynamically allocate the population size, the number of refinement parameters and the number of constraints. Thus one often may not have to use this command. If previously allocated arrays are reallocated, DIFFEV tries to save the old values and you can continue to use these. If the new array sizes are smaller than the previous ones, this can obviously not be done. DIFFEV will perform the new allocation, but all old data are lost. A short warning will be printed.

```
allocate
allocate "show"
```

Without parameter or with the parameter "show", the allocate command shows the current memory allocations.

```
allocate "default"
```

Allocates all array sizes to default values.

```
allocate "constraint", <max_constr>
```

Allocates the maximum number of constraints that DIFFEV shall handle.

```
allocate "population", <max_members>, <max_parameters>
```

Allocate the maximum population size and the maximum number of parameters that can be refined.

A.6 adapt

```
adapt "sigma" ,<parameter_number>,[,{"yes"|"no"|<value>}]
adapt "lsigma",<parameter_number>,[,{"yes"|"no"|<value>}]
```

DIFFEV uses to "sigmas" to handle special situations.
The global sigma pop_sig[<i>] is used in the following two situations: - a new parameter falls outside the range allowed by pop_xmin[<i>] and pop_xmax[<i>]. In this case the new parameter is chosen by adding a Gaussian random distributed value with sigma pop_sig[<i>] next to the respective boundary. - the difference between two parameters is zero. This will usually occur for integer parameters only. In this case the new parameter is chosen by adding a Gaussian random distributed value with sigma pop_sig[<i>] to the value of the effective donor.
If sigma is allowed to adapt during the fit, its value is set to (maximum parameter value - minimum parameter value) * <value> Thus, as the population converges to a smaller parameter spread, sigma dynamically becomes smaller as well.
The adaptation can be set for each parameter <parameter_number> separately.
The local sigma is used to modify a member not by differential evolution, but by a adding a local shift to the member. The local shift is a Gaussian distributed random value with sigma = <lsigma>. Whether DIFFEV uses this mode, is determined by the ==> variable diff_lo[1]. This gives the probability, that a given member will be modified by the local change instead of differential evolution.

A.7 backup

```
backup "NONE"
backup <input> [,<extension] , <output>
```

This command allows you to back up the current best solutions. Diffev expects that the current trial solutions are called inputfile.???? or inputfile.????.extension where "????" is a four digit integer number with leading zeros. Note that the '.' before the number "????" and before the extension are mandatory parts of the file name. You need to ensure that your version of "kup.diffev.mac" adheres to this standard. If the output file name includes a path, make sure that the output directory does exist. DIFFEV does not create the output directory.
Examples backup CALC/calc, FINAL/final This form will back up the files "CALC/calc.????" as "FINAL/final.????" backup TEMP/calc, tth, FINAL/final This form will back up the files "TEMP/calc.????.tth" as "FINAL/final.????.tth"
If multiple files need to be backed up, use the 'backup' command repeatedly:
backup TEMP/calc, tth, FINAL/final backup TEMP/calc, grcalc, FINAL/final These two lines will back up the files "TEMP/calc.????.tth" as "FINAL/final.????.tth" "TEMP/calc.????.grcalc" as "FINAL/final.????.grcalc"
backup NONE Turns off the back up option
The backup option is useful, if the calculation of the solutions takes a long time and involves random configurations. In these cases, the extra time required to copy the files may be well invested rather than to calculate these again after the end of the refinement. If the calculation

involves random configurations, a repeated calculation of the solutions on which the cost function depends may not yield exactly the same result. With the backup option you ensure that you always have those backed up solutions correspond to the actual cost function values that DIFFEV used.
If the calculation of the solutions is quick or if it does not involve random configurations it is faster not to run the backup during the refinements.

A.8 branch

```
branch kuplot [, "-macro" <macro_name> [ <par1> [ , <par2> ...]]]
branch discus [, "-macro" <macro_name> [ <par1> [ , <par2> ...]]]
```

Active within the discus suite only!
Branches to the "kuplot" or "discus" section.
Within this section any standard KUPLOT command can be given. The behavior of "kuplot" is essentially the same as in the stand alone version. Likewise for DISCUS.
The main use will branch to KUPLOT while the discus section is run via run_mpi from a DIFFEV slave.
Optionally the "-macro" qualifier instructs the suite to run the macro <macro_name> (with its optional parameters) before the interactive session is started.

A.9 compare

```
compare ["silent"]
```

This is the main part of the differential evolution section. This command reads the current results that the slave program stored in files ==> 'restrial' and compares these to the results of the parent generation. A new set of children is calculated according to the differential evolution algorithm. The successful parents are written to the file ==> 'logfile', which stores their respective cost function, and the full parameter set. At the same time the short summary file ==> 'summary' is appended with abbreviated information about the last generation. The new children parameters are written to the temporary files ==> 'trialfile'. The current generation is increased in file 'GENERATION'.
If the optional "silent" qualifier is specified, DIFFEV will not read the result files. Instead, DIFFEV must have received the current results by explicitly setting the values of child_val[*] for all children. This option will work only within the discus_suite, which is a collection of DIFFEV, DISCUS, and KUPLOT into a common program.

A.10 constraint

```
constraint <logical expression>
```

The parameter range may be restricted by defining one or several constraint conditions. Each condition must be a valid logical expression. For details of the syntax see the manual entry under ==> Command language. The parameters within the condition are referred to by p[<i>],

where <i> is parameter number, 1 up to the dimension of the problem at hand. The dimension is fixed through parameter ==> 'pop_dimx', see the variable entry.
Example

```
constraint  p[1].gt.2   # The first parameter must be larger than 2
constraint  4.le. p[2]**2 + p[3]**2
                        # The sum of parameters 2 and 3 squared must be
                        #  equal or greater than 4.
```

If a constraint equation is not met, DIFFEV will create a new parameter set. This process is repeated until a valid parameter set is found, or until DIFFEV has tried so for MAX_CONSTR_TRIAL times. In this latter case the program stops.

A.11 data

```
data "kuplot", <number>
data <type>, <infile>
data "kuplot", <number>
data <type>,<infile>
```

Currently the command serves toload the data set once and thus to place proper values into the system variables: F_XMIN = xmin[1] F_XMAX = xmax[1] F_XSTP = (xmax[1] - xmin[1]) / (np[1]-1)
Likewise for 2D data the variable F_YMIN, F_YMAX, F_YSTP are set. The loaded data set is currently discared.
Refer to the KUPLOT section for instructions on loading data.

A.12 deallocate

```
deallocate {"all" | "constraint" | "population"}
```

This command allows to deallocate memory for the specified program segments. This helps to conserve memory, if program sections are no longer needed.
This deallocation applies to memory that you allocated yourself and also to memory that DIFFEV has allocated automatically during runtime. As DIFFEV does not necessarily know, when you do not need the results of certain calculations any longer, it does not deallocate the automatically allocated memory sections unless you tell DIFFEV to do so.

```
"constraint"
```

Free memory associated to the maximum number of constraints.

```
"population"
```

Free memory associated to the population size and maximum number of refineable parameters.

A.13 dismiss

```
dismiss {<n> | "all" }
```

Set the R-value of the worst <n> parents to a very high value. Thus <n> of the next children will definitely have a better R-value and replace these parents.
This command should be used after you changed the parameter dimension and initialized ==> 'init' some or all of the trial values. Otherwise, many or eval all of the new children might not have a better R-value than their parents, and as a consequence the (re-)initialization of the trial values may get lost.

A.14 donor

```
donor {"best" | "random"}
```

The donor vector may be chosen in two different ways. "best" chooses the parent member that has the best parameter set. "random" chooses at random one of the parent vectors as donor vector. Two other parent vectors are always chosen at random to form the difference vector that is added to the donor vector. DIFFEV chooses the effective donor base along the straight line from the current parent vector to the donor. The point along this line is determined by the value of "diff_k[1]" (==> variables). For diff_k[1] = 0 the effective donor is the parent vector, for diff_k[1] = 1 the effective donor is the donor vector itself.

A.15 fix

```
fix <parameter_no>, {<value> | "best"}
fix <parameter_name>, {<value> | "best"}
```

This command fixes the value of parameter <parameter_no> or <parameter_name> to <value> for all members of the population. The ==> refine flag is turned of for this parameter. If the second parameter is "best", the parameter is set to the corresponding value of the current member with the best R-value.
As a side effect, the values of pop_xmin, pop_xmax, pop_smin, pop_smax are fixed to this new value as well.

A.16 functions

The following DISCUS specific functions exist. For a listing of general intrinsic functions see help entry 'functions' in the 'Command language' section of the online help.

```
par_number(<char_variable>)
```

Returns the number of the refinement parameter whose name is encoded in the character variable. Example: line = 'P_length' eval par_number(line)

```
par_name(<number>)
```

Returns the name of the refinement variable number <number>

A.17 initialize

```
initialize [ <par_number1> [, <par_number2>]] [, "silent"]
initialise "logfile"
```

This command initializes the differential evolution sequence.
Before using this command, you must have defined:

```
Number of parameters to be defined ==> 'pop_dimx'
Size of the parent population      ==> 'pop_n'
Size of the children population    ==> 'pop_c'
Boundaries for each parameter      ==> 'pop_xmin'
                                   ==> 'pop_xmax'
Starting intervals for parameters  ==> 'pop_smin'
                                   ==> 'pop_smax'
Sigmas for parameter adjustment    ==> 'pop_sig'
Local sigmas                       ==> 'pop_lsig'
Cross over probability             ==> 'diff_cr'
Fraction of the difference vector  ==> 'diff_f'
Point between parent an donor base ==> 'diff_k'
Local search probability           ==> 'diff_lo'
```

Without the optional parameters, 'initialize' is used to start the generation zero.
'Initialize' will use this information to generate the zero's generation. The file 'GENERATION' is set to generation zero, the population size and the number of parameters is written. The files ==> 'logfile' and 'summary' are initialized. Old versions with the same name are overwritten! The starting parameter values are written to files ==> 'trialfile' After the header, each line contains a one parameter, 'pop_n' (i.e. the size of the population) lines are written.
If you want to reinitialize one or several parameters, the 'initialize' command may be used with the optional parameter(s). In this case 'initialize' will simply set the corresponding parameter, or parameter range from <par_number1> to <par_number2> to the range defined by the values of pop_smin[*] and pop_smax[*]. A new set of children and the GENERATION file is written.
If the last parameter is the string "silent", the trial files are note written to your disk. DIFFEV expects to be part of a suite program and will transfer the trial parameters directly to the slave program. See also ==> 'trialfile', 'run_mpi', 'compare' This option will work only within the discus_suite, which is a collection of DIFFEV, DISCUS, and KUPLOT into a common program.
The second form of the command can be used to initialize just the logfiles, see ==> 'logfile', 'summary'. This might be helpful if you want to reset the refinement cycle to a smaller generation number. For very length refinements with a few thousand refinement cycles and many parameters, a continuation will take appreciable time to read all previous cycles. If the actual development across the cycles is not relevant to you you can reduces the file sizes drastically be a sequence like: pop_gen[1] = 1 initialize logfile

A.18 lastfile

```
lastfile <filename>
```

This defines the short log file of the parameter evolution.
After each generation, the short lastfile <filename> is overwritten by the parameters of the current generation.

It is a SPEC type file that contains all old R-values and parameters. Only this last generation makes up a scan. The first column is the member number, the second the R-value and all further columns the respective parameter values.

A.19 logfile

```
logfile <filename>
```

This defines the log file of the parameter evolution.
After each generation, the logfile <filename> is appended by the parameters of the current generation.
It is a SPEC type file that contains all old R-values and parameters. Each generation makes up a scan. The first column is the member number, the second the R-value and all further columns the respective parameter values.

A.20 newparam

```
newparam <name>, range:[<xmin>, <xmax>],value:[<smin>, <smax>]
         [, status:{"free"|"fixed"}
         [,init:{"keep"|"initialize"}
         [,type:{"real"|"integer"}
```

WARNING, non backward compatible change in 6.21.00
This command defines a new parameter name or changes the setting for an existing parameter name. range:[<xmin>, <xmax>] are the absolute lower and upper windows for the parameter. DIFFEV restricts the refinement to this interval. value:[<smin>, <smax>] are the lower and upper limit of the starting window that is used when a parameter is initialized.
status:free This variable shall be refined status:fixed This variable shall be kept fixed. DIFFEV will fixe the range and the value to the lower limit
The optional parameter "init" defines if the current parameter values for the population are kept at their current values or if this parameter shall be initialized within the start window <smin>, <smax>.
The optional parameter "type" defines if the current parameter shall be treated as a real valued number (the default) or as a whole, integer number.
The command summarizes the individual commands par_name <name> pop_xmin[<ipar>] = <xmin> pop_xmax[<ipar>] = <xmax> pop_smin[<ipar>] = <smin> pop_smax[<ipar>] = <smax> pop_sig[<ipar>] = 0.001 pop_lsig[<ipar>] = 0.0001 type {"real"<"integer"}.<ipar>

A.21 purge

```
purge <n>
```

This command removes all info on the initial <n> generations from the logfiles in folder DIFFEV. This will significantly shorten the log files in case of a lengthy DIFFEV refinement.

A.22 read

```
read
```

This command has been removed. Activated again in version 5.29.0
The read command instructs DIFFEV to read the GENERATION file and to determine the population from the Parameter files.
This command is only needed if you want to add a new parameter to a refinement that you want to continue and the ==> 'newparam' command is the first command in DIFFEV after starting the program prior to the ==> 'run_mpi' command.
Example: read newpara P_new, 0.0, 10.0, 10.0, 9.0 run_mpi discus, dis.diffev.mac, repeat:5, compute:parallel compare
The 'read' command is not necessary if now new parameters are defined. In this case, the 'run_mpi' command will automatically determine the refinement state from 'GENERATION'

A.23 refine

```
refine {"all"|"none"|<number> [,<number>...]}
refine {"all"|"none"|<par_name> [,<par_name>...]}
```

The refine command is obsolete, you will get a warning message if the command is used. Check the 'newpara' command for the new style to fix/free a parameter.

A.24 release

```
release {<par_name>| <number>} , range:<sigma>
  [value:<setpoint>]
  [min:<pop_xmin>]
  [max:<pop_xmax>]
  [dismiss:<number>]
  [dismiss:"all"]
  [dismiss:"best"]
  [dismiss:"none"]
```

This command will initialize a parameter named <par_name> to a range around the current best value. Its intent is to act complementary to a ==> 'fix' command. It works in a similar fashion as the ==> 'init' command, but the user does not have to ensure that the ranges of pop_xmin to pop_xmax and pop_smin to pop_smax are non-zero.
The parameter will be refined, i.e. the 'release' command implies a ==> refine <par_name> command.
If the optional parameter 'value:<setpoint>' is present, it defines the setpoint around which the parameter values will be initialized. If the optional parameter is omitted, the setpoint defaults to the value of the current best population member. The initialization range is set to the setpoint +- range.
The absolute limits, ==> 'pop_xmin', 'pop_xmax' are set to a range of +- 3* <range> around the <setpoint>. The original user provided values of pop_min or pop_max are replaced by the new values.

If you need different ranges for pop_xmin and pop_xmax, or if the new limits might be outside a physical limit you need to set the proper limits explicitly with the optional parameters "min" and "max": min:pop_xmin max:pop_xmax
If all the children will produce R-values that are worse than the parents, DIFFEV will discard these children and create the next generation based on all parents. As all parents reflect the state prior to the release, they all correspond to identical values of the parameter that you wanted to release. In effect, the release is lost. To prevent this, the 'dismiss:' option will set the R-value of a user specified number of parents to a very large value. This ensures that the corresponding children will certainly survive into the next generation. The values that the 'dismiss:' parameter may take are: <number> : A number from 0 to the population size pop_n[1] The <number> worst parents are dismissed. "all" : All members are dismissed. "best" : All but the best member are dismissed. "none" : No member is dismissed. Same as <number>=0.

A.25 reset

```
reset
```

Resets DIIFEV to the conditions at program start. The generation number is set to zero, the population size, children size and the number of parameters is set to the default value of one. All parameter names are removed from the list of user variables.
Use this command if you want to combine several refinements for different experimental data to be executed one after the other. Without the reset, the old parameters would persist and may interfere with your new parameters.

A.26 restart

```
restart <user_generation>
```

If you set <user_generation> to a value that is less than the current value in GENERATION, the Logfiles, Summary and Last files are shortened and the GENERATION file is adapted. This lets you step back to an earlier refinement status. Keep in mind that the later cycles are irrevocably lost. If in doubt make a backup first.

A.27 show

```
show population
show parameter
show config
```

population Shows information on the current population.
parameter Lists all parameter names their limits and current range.
config Current maximum values, these are overwritten dynamically.

A.28 summary

```
summary <filename>
```

This defines the log file of the R-value/cost function evolution.
The SPEC type summary files contains a single scan. Each generation creates one line within the scan. The first columns is the generation number. For the R-value and each of the parameters four columns are written. The first of these is the average value, the second the minimum value, the third the maximum value and the fourth the sigma of the parameter distribution.

A.29 restrial

```
restrial "silent" {,partial:<no>}
restrial <result> {,partial:<no>}
```

Within the discus_suite always use the "silent" mode!
These temporary files are used to communicate the R-value from the slave program back to DIFFEV. DIFFEV expects a separate file for each member of the population. The file name <result> is automatically augmented by a four digit member number. Thus, if the file name is Results, DIFFEV expects to find the files:

```
Results.0001
Results.0002
etc.
```

The file contains one line with two numbers, the member number and the R-value obtained for the corresponding set of parameters in file ==> 'trialfile'.
If the file name is the string "silent", the result files are note read from your disk. DIFFEV expects to be part of a suite program and will expect that the result values have been transferred directly from the slave program. See also ==> 'trialfile', 'run_mpi', 'compare'
The optional parameter "partial:<no>" tells DIFFEV how many partial R-values to expect from the slave program. Within DIFFEV the partial R-values are just archived, the slave program has to provide the properly weighted average R-value.

A.30 run_mpi

```
run_mpi <program_name>, <macro_name>, <no_repetitions>, <output_base>
```

This command starts the processing of the slave program <program_name>. If the program has been compiled with MPI and started with mpiexec, a parallel computation is started. Otherwise the slave program is executed in a serial loop over all children and individual repetitions. This command starts parallel processing of program <program_name>. The program will be executed <pop_c>*<no_repetitions> times. <pop_c> is the number of children for the current refinement and each child corresponds to one ==> trialfile. The calculations can be repeated <no_repetitions> times, with the identical parameter set. This may be necessary, if you need to average several calculations.

MPI option: The standard output of the program will be directed into a file <output_base>. The current child number (and the current repeat number, if present) will be appended as four digit wide field. On a UNIX system you can redirect the output to /dev/null.
NON MPI option: Regardless of the value of <output_base>, the output is displayed at the screen. The usual options for the ==> 'set prompt,' command do hold.
Starting with version 5.7 the communication between the diffev section and the discus or kuplot section is done via internal variables. These are: REF_GENERATION : the current generation number REF_MEMBER : the number of members in the population REF_CHILDREN : the number of children in the population REF_DIMENSION : the number of parameters defined in DIFFEV REF_KID : the current child REF_INDIV : the current repetition for the current child REF_KID REF_NINDIV : the intended total number of repetitions
The actual parameters are transferred to DISCUS/KUPLOT via variables ref_para[...] : Indices from 1 to REF_DIMENSION
The capitalization of the refinement variables is intended to distinguish these from user variables. Other than that they should be used just as any user defined variable, with no square brackets: kid = REF_KID
IN a similar fashion, the KUPLOT rvalue command transfers its result back to DIFFEV internally. See 'resfile silent'
Essentially, DIFFEV branches to the section, executes the macro file and returns to DIFFEV. As the communication is done via the internal variables, the need for the macro parameters included in previous version 5.5 and earlier has ceased. You are encouraged not to rely on macro parameters but to use the refinement variables.
Relevant for older verisons (5.6 an earlier only):
The actual line that starts the program has syntax: "program_name" -macro <macro.name> <cwd> <child> <indiv> > <output>
<macro_name> name of the macro file to be executed <cwd> 1st macro parameter string with current directory <child> current child number (1 to pop_c) <indiv> current repetition number (1 to <no_repetitions>) This parameter is omitted, if <no_repetitions> is zero <output> File for standard output written by <program_name>
Example run_mpi discus, discus.mac, 5, LOGFILES/d Runs discus with macro "discus.mac". For each child the calculation is repeated 5 times. DISCUS output is written into files "LOGFILES/d.xxxx.yyyy", where "xxxx" is the child number and "yyyy" the repetition number. run_mpi discus, discus.mac, 5, /dev/null Same, except that the output is written to "/dev/null" i.e. it is thrown away. run_mpi discus, discus.mac, 0, LOGFILES/d Same, except that no repetitions are requested. Only one calculation is performed per child and the output is written to "LOGFILES/d.xxxx".

A.31 selection

```
selection {"compare" | "best","all"}
```

This command governs the selection criterion that determines which children and parents survive into the next generation.

```
"compare"    Each child is compared to its immediate parent. The better
             of these two will survive into the next generation. It will
```

```
             serve as one of the parents from which the next children
             are derived using the Differential Evolution Algorithm.
             The number of children ==> 'variables' pop_c, should be
             identical to the number of members i.e. parents
             ==> 'variables' pop_n.
"best","all" All parents and all children are put into a single list.
             Of this list the best <pop_n[1]> members survive into
             the next generation. No restriction applies to keep any
             parent or any children.
             You are free to use any number of children. If the number of
             children is increased compared to the number of parents, the
             selection becomes "tougher" since a smaller percentage of the
             whole population survives into the next generation. This will
             speed up convergence, yet run a higher risk of getting stuck
             in a local minimum.
```

A.32 trialfile

```
trialfile <filename>|"silent"
```

These temporary files are used to communicate the current set of parameters between DIFFEV and the slave program.
DIFFEV writes a separate file for each member of the population. The file name <filename> is automatically augmented by a four digit member number. The if the file name is "Trials", DIFFEV expects to find the files:

```
Trials.0001
Trials.0002
etc.
```

The file has format:

```
# generation members children parameters
     181        45        90         4
# current member
    1
# parameter list
    0.8284027688E-02
    0.5815573883E+02
    0.2050311089E+01
    0.3000000000E+01
```

The first states the current generation, the number of members in the population, the number of children in each population, and the number of parameters. The next two lines states the number of the current member. This is followed by a list of all parameter values, each in a separate line.
If the file name is the string "silent", the trial files are note written to your disk. DIFFEV expects to be part of a suite program and will transfer the trial parameters directly to the slave program. See also ==> 'trialfile', 'run_mpi', 'compare' This option will work only within the discus_suite, which is a collection of DIFFEV, DISCUS, and KUPLOT into a common program.

A.33 type

```
type {"integer" | "real"},<number>
```

Defines the number that the parameter <number> assumes. Valid options are:

```
"integer" The parameter is restricted to integer numbers.
"real"    The parameter may take on any real,floating number.
```

See the entry on 'variables' regarding options to limit the allowed range, and the entry ==> 'constraint' on possible constraints.

A.34 variables

Like all programs of the Diffuse suite, diffev offers integer and real variable for standard calculations == Command Language/variables
Unique diffev variables are:

```
pop_gen[1]     The number of the current generation
pop_n[1]       The size of the parent population
               The size of the population may be changed during a
               refinement. In this case the GENERATION file is updated
               automatically. If the population is increased, the R-values
               for the new members are set to ten times the maximum
               current R-value. Parameter values are copied from old
               member number one.
pop_c[1]       The size of the children population
               The size of the children population may be changed during a
               refinement. In this case, a new set of trial files is
               automatically generated and the GENERATION file is updated
               as well.
pop_dimx[1]    The dimension, i.e. the number of parameters
               The dimension may be increased during a refinement. To do
               so you need to increase pop_dimx[1], set all parameter
               related values for this new parameter and run the
               ==> 'init <par_number>' command. This will initialize the
               parameter to the starting range and update the trial files
               and the generation file.

pop_xmin[<i>] Minimum allowed value for parameter <i>
pop_xmax[<i>] Maximum allowed value for parameter <i>

pop_smin[<i>] Minimum starting value for parameter <i>
pop_smax[<i>] Maximum starting value for parameter <i>

pop_sig[<i>]  Sigma of Gaussian distribution for parameter <i>. If the
               difference between two parent parameters is zero, or if a
               child parameter is outside the limits defined by pop_xmin
               and pop_xmax, the corresponding parameter is modified by
               a Gaussian distributed random number.
               Set pop_sig to zero to switch off this option.

pop_lsig[<i>] Sigma of local Gaussian distribution for parameter <i>.
               The probability diff_lo[1] determines if a given member
               is changed locally only or takes part in the usual
               differential evolution algorithm. If it is changed only
               locally, the parent parameters are modified by adding
               a Gaussian distributed random number, with mean zero
               and sigma pop_lsig.
```

```
pop_v[<i>,<j>] Value of parameter <i> for member <j>
               This parameter is read only.

pop_t[<i>,<j>] Current trial value of parameter <i> for child <j>

rvalue[<i>]    R-value for member <j>. This is the R-value for the
               current parent generation.

bestm[1]       Parent member that currently has the lowest R-value

bestr[1]       Currently lowest R-value.

worstm[1]      Parent member that currently has the highest R-value

worstr[1]      Currently highest R-value.

p[<i>]         Parameter symbol used in the constraint conditions.

diff_cr[1]     Cross over probability
diff_f[1]      Multiplier for difference vector
diff_k[1]      Multiplier for vector between parent vector and donor
diff_lo[1]     Probability for local refinement of a population member
diff_sel[1]    Selection mode for compare command, READ_ONLY
```

A.35 write

```
write {"children" | "generation"}
write "children"
```

The write command will generate a new set of child values and update the corresponding trial files and the GENERATION file. The generation number is not changed.

```
write "generation"
```

Writes the GENERATION file. No changes are done to the file.

Bibliography

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